Plants as a Structural and Material System

A dissertation examining plants as both a structural and material system to reduce carbon emissions and deliver socio-economic change. Hemp (Cannabis Sativa L) is assessed as a building material, alongside Frei Otto's tree-branching structural systems, to inform a more sustainable, renewable approach to construction.



Case studies of branching structural systems and hempcrete as a material system are combined to test whether plant-based construction can deliver a sustainable, renewable building with improved performance and reduced environmental impact.

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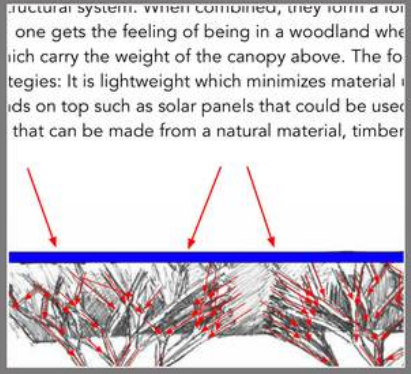


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Plants as a Structural and Material System

Hemp as a Structural and Material System

This dissertation, Plants as a Structural and Material System, examines how plants can reduce carbon dioxide emissions and provide socio-economic change in construction. Trees offer a natural precedent for structural systems; research into Frei Otto's branching structures and the BIC value method is used to optimise support and minimise material use. Hemp is examined as a material system, assessed through an interview with UK Hempcrete and precedents including the WISE building, to understand how hempcrete can form the non-load-bearing fabric of a building. Physical hanging models and hempcrete material tests were used to explore how a branching structural system and a hempcrete material system could work together, directing water and dispersing structural force while minimising material usage. The dissertation concludes that combining a plant-derived structural system with a plant-derived material system can produce a sustainable, renewable building with improved performance and reduced environmental damage.

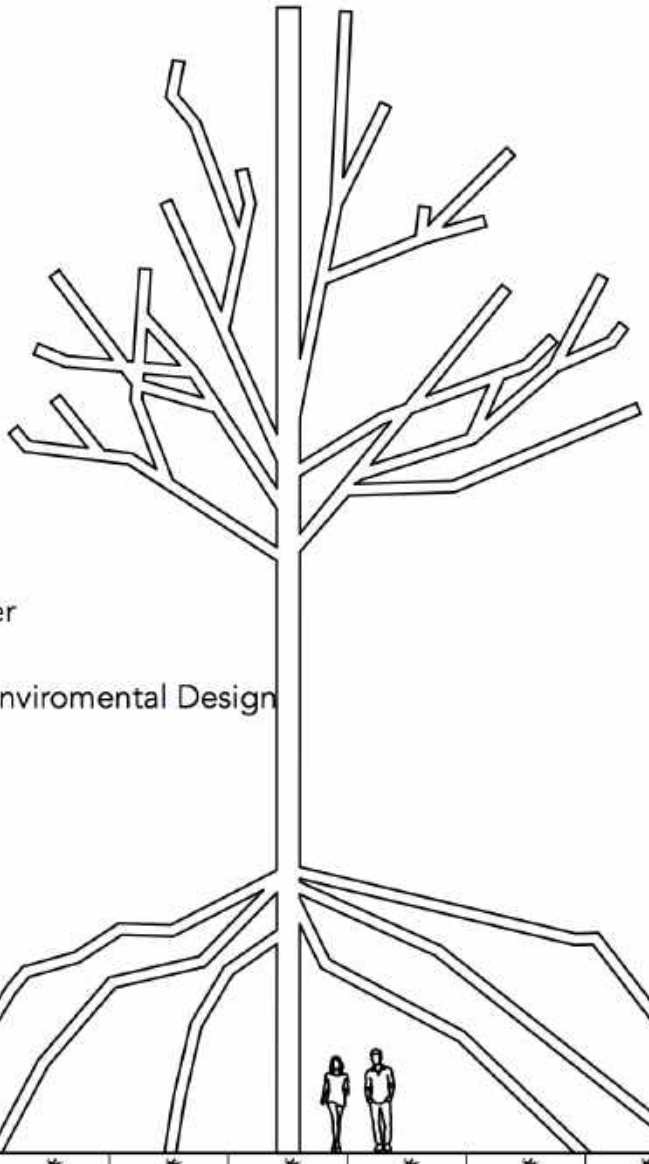






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Plants as a: structural and material system



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Abbreviations

3D- Three dimensional [1]

THC-Tetrahydrocannabinol[2]

IPCC-Intergovernmental Panel on Climate Change[3]

UNEP: United Nations Environment Programme

LCDA- La Chanvrière de l'Aube[4]

CAD- Computer aided design[5]

CNC-Computer numerical control[6]

THC-Tetrahydrocannabinol[7]

IPCC-Intergovernmental Panel on Climate Change[8]

CO2- Carbon dioxide[9]

HVAC- Heating ventilation and air-conditioning[10]

CAT- Centre for alternative technology[11]

FSC- Forest Stewardship Council[12]

CRS- Congressional Research service[13]

[1] (Otto & Rasch, 1995)

[2] (Herer, 2010)

[3] (UNEP, 2011)

[4] (Abel, 1980)

[5] (John Fraser, 1995)

[6] (Hudert & Pfeiffer, 2019)

[7] (Herer, 2010)

[8] (UNEP, 2011)

[9] (IPCC, 2018)

[10] (UNEP, 2011)

[11] (CAT, 2019)

[12] (CAT, 2019)

[13] (CRS, 2019)

Plants as a structural and material system; to reduce carbon dioxide emissions and provide socio-economic change. Plants can create buildings that reduce carbon dioxide emissions, sourced from nature; they give life to self-sustaining renewable construction materials [14] Tools from the past and present must be used to fit the current modern-day needs for modern-day solutions. The building sector, which is a primary contributor to the negative environmental impacts, needs to be re-designed, ways of the past are no more, and solutions to redirect the course of humanity are required [15] [16]. Using new mind-sets and resources, we need to evolve the way we design and construct buildings.

. "Return of nature" (Hudert & Pfeiffer, 2019, p. 180).

Plants can form structural elements[17][18]. As inspiration, they lead to unique, diverse forms, intricate in their design[19]. Through research and studies a system to minimize material use and maximize performance can be developed, a structural system to reduce environmental damage [20] [21]. As a structural element alone is not enough to form an entire building, a plant called Cannabis Sativa L – Hemp, as a building material system, can be used to create the non-load bearing aspects of a building [22] . An online interview with UK Hempcrete who uses Tradical Hempcrete (Tradical 2019) was conducted to assess whether hemp can be utilized as a material system (Windsor,2019). A structural system developed by an Architect, Frei Otto; and the WISE building which shows the use of hemp as a material system, will contextualize the idea put forward and show how each system works in a real-life context.

The research will show how nature, more specifically plants can create buildings which bring environmental and socio-economic benefits to the global economy. Changes in perceptions and new ways of working have led to innovations seeping into modern-day construction; these changes are crucial to understanding how plants can assist in rectifying the current environmental issues and bringing socio-economic change [23] [24]. The conclusion will lead to the start of a thesis project that aims to assess whether the structural system, when combined with the material system, leads to a sustainable renewable building with increased performance benefits, and less environmental damage.

[14] (Herer, 2010)
[15] (IPCC, 2018)
[16] (UNEP, 2011)
[17] (Otto & Rasch, 1995)
[18] (Hudert & Pfeiffer, 2019)
[19] (Forshaw, 2016)
[20] (Otto & Rasch, 1995)
[21] (Nerdinger, 2005)
[22] (Windsor, 2019)
[23] (Johnson, 2018)
[24] (Michael H. Ramage, 2016)

A plant, a tree, a flower; using nature as a model to generate architectural systems, and bring environmental and socio-economic change. A recent report written by the Intergovernmental Panel on Climate Change (IPCC) shows significant differences in climate characteristics between the present and past with a 1.5°C and 2°C increase compared to pre-industrial levels [25]. Further expanded in United Nations Environment Program (UNEP) Green Economy report, which provides explanations to the problems and solutions, thus; the need to substitute clean energy and low-carbon technologies for fossil fuels has become vital [26]. Human-made structures are incomplete and straightforward; they need to be re-designed to solve the ever-increasing environmental issues [27] [28]. Door-ways to a 'green' sustainable world are now arising as we evolve as a species both physically and psychologically [29] [30]. "Most cost-effective and efficient policies rely on enforcement of sustainable building standards" (UNEP, 2011, p. 29); standards to reduce environmental damage and produce a sustainable and renewable building sector.



Figure.....20:Model of the structural system

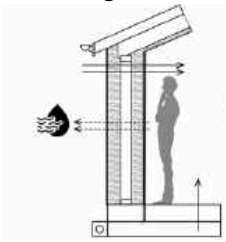


Figure.....47

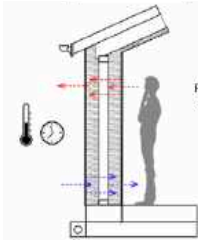


Figure.....48

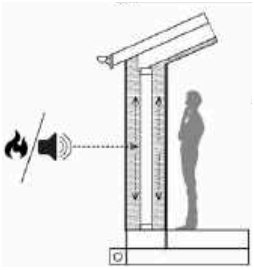


Figure.....49

Plants as a structural and material system; to reduce carbon dioxide emission and provide socio-economic change may be a tool to construct such change. The extent to which plants can be used as inspiration and application for form and function, to bring positive environmental and socio-economic benefits to the global population will be explored [31] [32] [33]. Nature as a paradigm for architectural design [34]; analysing this to create an understanding of how using nature as inspiration for design may help solve current issues [35]. To form a building, structural and material systems are needed. When observed, plants as a structural system behave uniquely. When transformed through a mechanical view, the result is a lightweight system that maximizes performance, a plant-based *structural system*[36] [37] [38]. Hemp was once used worldwide but with rising concerns on its varied uses, became illegal in the 1930s, since the 1990s new laws have led to its rediscovery and when combined with structural elements, it gives life to several uses and performance benefits; a plant-based *material system* [39] [40] [41] [42]. These systems will be linked through the environmental and socio-economic benefits they provide and their use as a strategy, as highlighted in the Green economy report, to reduce the negative impacts that are currently facing planet earth; explicitly looking at carbon dioxide reductions (CO_2)[43]. Concluding to find if one may say--Plants as a structural and material system; can reduce carbon dioxide emissions and provide socio-economic change.

[25] (IPCC, 2018)
[26] (UNEP, 2011)
[27] (Otto & Rasch, 1995)
[28] (John Fraser, 1995)
[29] (Otto & Rasch, 1995)
[30] (Stanwix & Sparrow, 2014)
[31] (Forshaw, 2016)
[32] (Otto & Rasch, 1995)
[33] (Stanwix & Sparrow, 2014)
[34] (Forshaw, 2016)
[35] (Forshaw, 2016)
[36] (Otto & Rasch, 1995)
[37] (Nerdinger, 2005)
[38] (Hudert & Pfeiffer, 2019)
[39] (Herer, 2010)
[40] (Boyce, 1900)
[41] (Dodd, 1876)
[42] (Stanwix & Sparrow, 2014)
[43] (IPCC, 2018)

Using nature as inspiration for design may lead to new diverse buildings to solve current issues. Art Nouveau, planted the seeds that can now bear fruit, by changing ones' perception of how nature can be incorporated into design, as stated by Forshaw, it is "Desire to create intricate architecture" (Forshaw, 2016, p. 17)(Appendix A: figure 6). A vision of what is needed now, a change in mind-sets and approaches to design, *Art nouveau*, a movement seen as an antidote against the constraints of Neoclassicism; a combined formula of mechanical and academic processes of the industrial revolution. Based on forms found in nature such as flowers and plants, the sensuous curves were a reaction against linear design and mass bland production. Using steel and glass, the movement took architecture on a new pathway that must be re-visited to repair our current problems, nature to drive design, re-thinking design to suit current needs, biometric design, using nature as a design driver[44]. An Art Nouveau pioneer, Victor Horta[45], redirected the course of architecture using the latest technology and nature to inform design against the norms of the time. Together with Paul Hankar[46], Horta created a movement into the elysian creature that was- Art Nouveau. Changing the way people saw buildings and how architects saw nature, using shapes and forms derived from nature.



Figure…..6: L'église de la Madeleine
Neoclassicism



Figure…..7: Basilica de la Sagrada Familia
Art Nouveau

The idea was nothing new; inspiration came from that which was here before - Nature. The movement picked up amongst high ranking officials and the cultural elite, leading to a worldwide trend from Europe to America. Sadly, it lost its trend due to its expensive nature and intricate design, as at the time, art deco and modernism rose, with it, the fall of art nouveau [47]. To this day the Basilica de la Sagrada Familia, designed and started by Antoni Gaudi in Barcelona is still being completed (Appendix A: figure 5,6,7). The movement created new ideas of how nature could be used as inspiration; diverse, intricate and never the same Art nouveau architecture is driven by nature, to inspire architects to create buildings that were not of the norm — a revolt against what had been done before. The lesson learnt from Art nouveau was to use nature to inspire design using the latest technology effectively. Art nouveau architects used nature as inspiration to challenge construction; architects now need to follow similarly, using the latest technology to create buildings that are not of the norm, to fix the negative impact that the building sector has placed on the environment [48].

[44] (Forshaw, 2016)
[45] Appendix A: figure 1&2
[46] Appendix A: figure 3&4
[47] (Forshaw, 2016)
[48] (Forshaw, 2016)

Systems derived from nature can reduce a building's environmental impact. A change in design to change the world, starting with the rise of Art nouveau in architecture, biometric design. A well-known architect, Frei Otto, was one of the decedents of biometric design. Constructing buildings in harmony with nature rather than in its opposition, "Architects to improve life… reverse catastrophic damage" (Otto, 2003). He viewed nature as necessary to the form-finding process in architectural research, using it to form designs [49]. Viewing nature as a natural process that creates objects, minimizing material usage and maximizing natural processes in form creation, resulting in natural construction [50]."Natural constructions are not concerned with the infinite possibilities of objects… rather concerned with those constructions which demonstrate the physical, biological and technical processes which objects give rise to" Frei Otto (Otto & Rasch, Finding Form, 1995, p. 23). Architecture to be aware of its creations, where the creators use design to produce complex objects, buildings with a purpose in form and function. Humans affect nature, so nature must be used to produce such objects, to control it in a way that produces unique outcomes. Humanity has been destroying nature since the beginning of the industrial revolution; we are now looking for ways of preserving it, to be a part of nature, a part of a whole, his means is technology which is compatible with nature[5152], using nature to derive a structural system may be the solution.

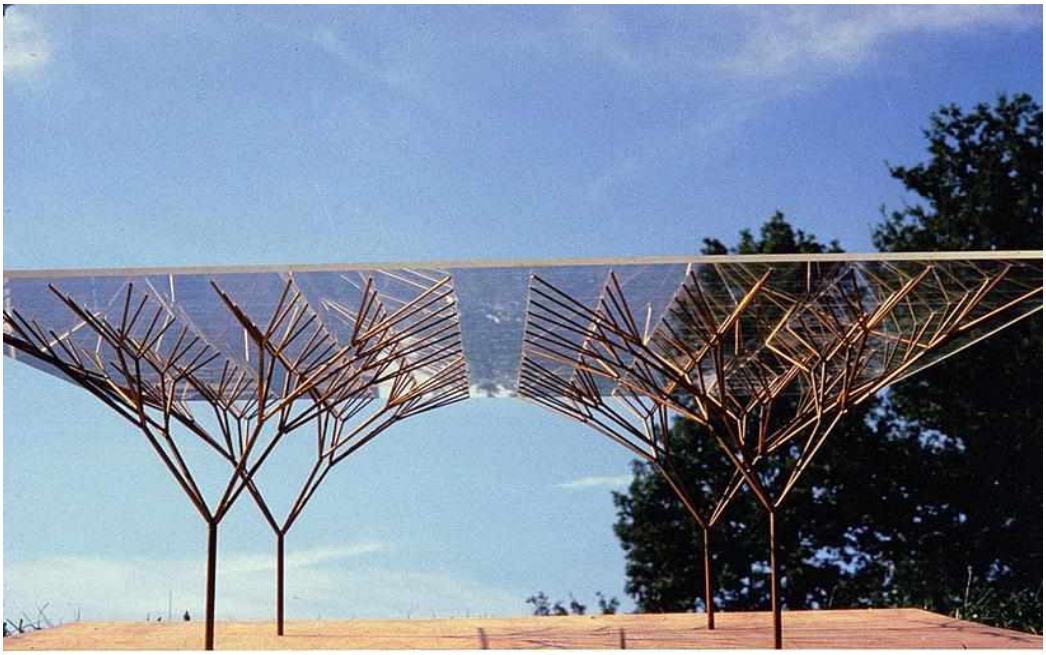


Figure.....20
Model of the structural system

[49] (Nerdinger, 2005)
[50] (Otto & Rasch, 1995)
[51] (Otto & Rasch, 1995)
[52] (Nerdinger, 2005)

Nature to produce architecture, a solution to current environmental problems. "Biology has become indispensable for architecture" (Otto & Rasch, Finding Form, 1995, p. 40); an idea of Frei Otto which led to buildings that were different and unique. Tackling issues of how man has made unnatural buildings, of which the times demanded, a re-call, to that which has been here since day one, nature. To seed design, sprouting it into lightweight, energy-saving, mobile, and adaptable buildings; natural buildings in short[53]. Using the technology of the time to aid and enhance design just at Victor Horta once did with steel, using it in his form-finding processes, with nature as a design driver [54]. Otto pursued the idea of developing models and methods in which forms generate themselves. Observing and analysing the processes, then creating new ideas from them: soap film experiments for tension loaded membranes, chains and nets to find forms for suspended construction and experiments with tilting and rotating discs to investigate the stability of masonry constructions when tilted or faced with an earthquake[5556] (as seen in appendix B Figures: 10 to 16)



Figure.....12:Soap film experiments

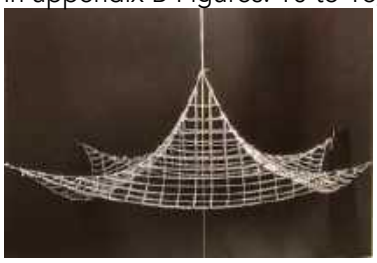


Figure.....14: Chain model



Figure.....16: Stability test



Figure.....19: Hanging model

Frei Otto was one of the first architects to challenge the misconception of nature; his various studies of tree branching systems derived from the forces of nature is an example of how nature can form a structural system. His ideas were revolutionary, giving views to new ways of approaching life, not only architecture. He brought topics together that were once unimaginable, as archetypes of the past did not reflect the changing demands of society; he adjusted accordingly [5758]. His branching tree structures were one of his designs that stood out, using natural forces and concepts to create branching systems that could bear loads while resembling natural tree systems [59]. This system can be incorporated into design to reduce material wastage and maximize the performance of supporting top-loads, inspired by trees to replicate their form and function to produce a structure that can be used in sustainable construction (represented in appendix B figures 20 to 22).

[53] (Nerdinger, 2005)
[54] (Forshaw, 2016)
[55] (Otto & Rasch, 1995)
[56] (Nerdinger, 2005)
[57] (Otto & Rasch, 1995)
[58] (John Fraser, 1995)
[59] (Nerdinger, 2005)

Case study: Tree branching system by Frei Otto and others

Trees as a structural system. Frei Otto's well-known branching system, created in 1960, at the University of Yale may be used as a structural system. The system resembles tree-like structures that could support loads based on calculations that would maximize the support and minimize the weight as well as material usage, using the BIC value and relative slenderness formula, diagrams, and experiments to provide a synopsis of all possible objects from a mechanical view. The BIC value is the relation of mass construction to its ability to transfer force over distance; the relative slenderness formula compares two objects looking at the relationship of distance of load transfer to force being transferred, the two formulas maximize the performance and minimize material usage .[6061] These formulas were used to asses different material and structural forms to derive the best material and way of constructing to result in an optimized design with maximum support load and minimal material usage (Shown in appendix B figures 17 and 18).

$$\lambda = s/\sqrt{F}$$

Figure.....17: Relative slenderness

$$BIC = m / Tra$$

Figure.....17: BIC value formula

Understanding that trees, naturally, are under bending stress, Otto wanted to eliminate this through a hanging model, using gravity to optimize the design for a single load scheme. Together with the formulas, he used models to experiment to find the best possible design; this was the start of the form-finding process for the branching system. The branching system creates various outcomes: a rooftop, in which loads are carried, creates a free open space underneath, shaded, and covered by the load-bearing structural system. When combined, they form a forest of trees, when walking through the space, one gets the feeling of being in a woodland where one is surrounded by branching trees which carry the weight of the canopy above. The forest of trees has several environmental strategies: It is lightweight which minimizes material usage, it is a shading device, it can carry heavy loads on top such as solar panels that could be used to harvest solar energy, and it is a structural system that can be made from a natural material, timber[62]

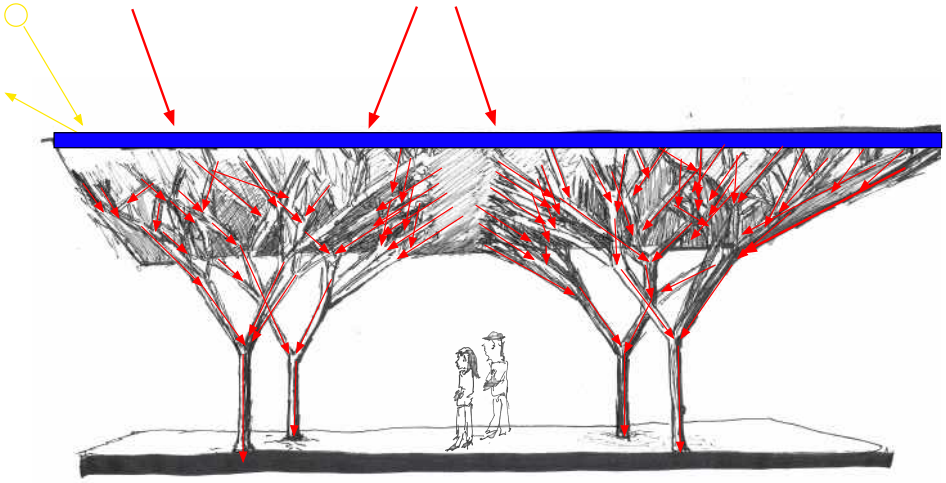


Figure.....21: Drawing of the structural system to show open space, shading and force being dispersed

[60] (Nerdinger, 2005)

[61]

[62] (Otto & Rasch,1995)

The system developed into a solution to minimize material usage and maximum load-bearing force, (drawn in appendix B figure 23). The formation of the system will allow one to understand how it can be used as a structural system. Trees are naturally under bending stress, but Otto wanted to eliminate this through a hanging model, using gravity and formulas to optimize the design for a single load scheme. A load-bearing scheme that spreads the force out across a distance to support the points; it is the flow of force in relation to shape, similar to "waters force through its structured leaves"(Otto & Rasch, 1995). Using branches to disperse the weight load, just as a tree does in the ramification of its non-wood parts that disperse weight load and carry force. Plants apply forces to many short-lived solar collectors and liquid routes both needed for survival to transport the energy for growth. Frei Otto's inspiration was derived from nature, as he was interested in how plants transport forces, he developed the system to carry force, but in his design, it was a load-bearing force.

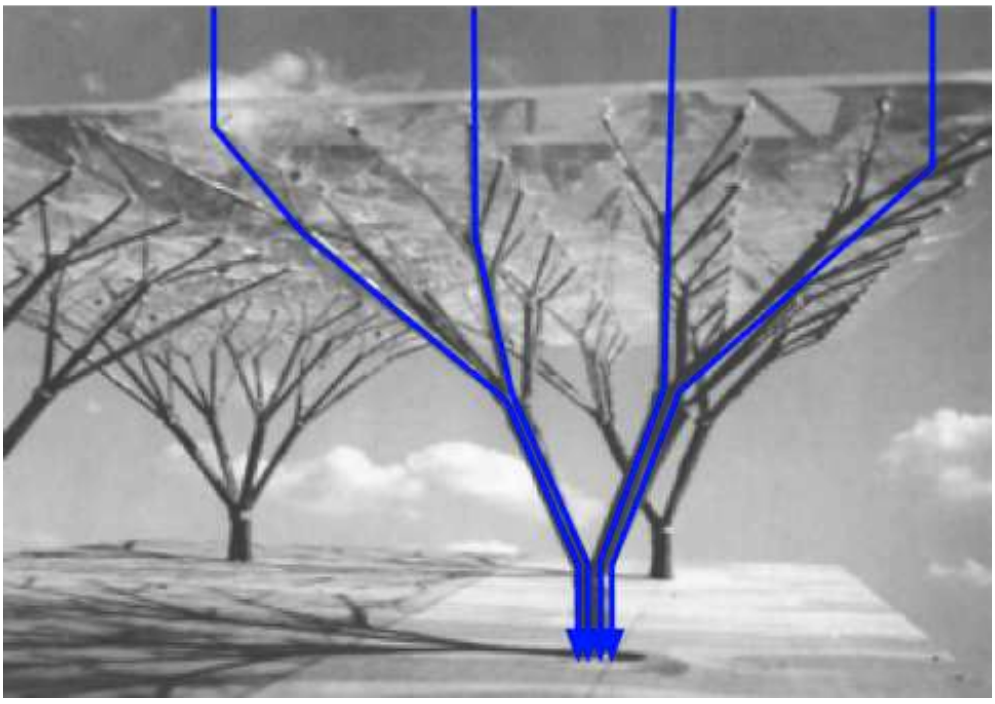


Figure.....9: Diagram of force load



Figure.....22: Drawing of leaf structure with water forces

Avant-garde in his approach to design, one can see how the master -Frei Otto- used nature to alleviate issues that were prevalent in construction at the time. Using nature to form design; the forces within plants are minimized for optimal growth[63]. The branching system uses nature to derive form, a structural system. This system produces a plant-based design that can be constructed from timber, a plant as a structural system to minimize material usage and maximize performance. As a system alone, it needs to be combined with a material system to complete the make-up of the building, a natural, sustainable material system. Hemp produces a system, when combined with the structural system, as mentioned, forming one whole renewable, sustainable building [64].

[63] (Otto & Rasch, 1995)
[64] (Otto & Rasch, 1995)

Hemp, in the past, was used in construction. Dating back to the 6th century, Hemp was used in mortar, discovered in Merovingian bridges dating back to 600AD (figure 24) [65]. It was first used as a building material in France. In 1958, Charles Rasetti set out to improve traditional construction. At the time, a mechanical process was being used to breakdown hemp plants by a local hemp farmers Co-Op company called La Chanvrière de l'Aube (LCDA)[66] [67]. Together with the local hemp supplier, Rasetti started to find the alternative uses of hemp; at the time, it was used mainly for paper and rope [68]. In 1986 "Canobioate" was patented by Rasetti and the LCDA; they registered, Canobioate, a process that coats the hurds with silica. A worker at the LCDA, Madame France Périer, patented a similar product called Isochanvre, there were no differences between the two materials apart from price with her Isochanvre being the most expensive. In the same period, a second approach was put forward by Yves Kühn, the difference in his approach is that he used untreated hurds and proposed using hemp hurds as a raw material mixed in with a lime plaster, creating "Canomose" or what is now known as 'hempcrete', an eco-construction material (figures 25 & 26)[69]. Once used to construct a bridge and then later developed into a building material system to complete the first house made with Hempcrete.



To this day both the bridge and Yves' house still stand, not only does hemp form a material for a material system but it is durable and can last for centuries, a truly sustainable material.

Figure.....27: First home made from hemp

[65] (Boyce, 1900)
[66] (Campos, 2016)
[67] (Roulac, 1997)
[68] (Lyster H Dewey, 1916)
[69] (Herer, 2010)

New laws now allow for hemp to be used as a material system. Currently, 30 countries produce hemp worldwide. In 1970 the Comprehensive Drug abuse prevention and control act was signed, linking all forms of cannabis together, which lead to hemp being classified as a psychoactive plant (figure28)[70] [71]. Due to new research and studies, the classification of hemp is now non-psychoactive. Laws have been put in place to distinguish what separates the two. The classification lies within the cannabidiol, the chemicals in cannabis: delta-9 Tetrahydrocannabinol (THC), the psychoactive quality of cannabis. A worldwide range of 0.2 THC to 0.3 THC defines hemp as non-psychoactive, which since the early 2000s has led to many countries changing their laws to suit this desired non-psychoactive range[72] [73]. This process has evolved through advancements in technology.

Advancements in technology led to evolution of hemp as a material system. In the past hemp was manually processed by hand. The traditional process is as follows: retting to separate the plants fibres, the Xylem (figure 35) from the skin by rotting away the cellulose that joins the fibres, it is then left to dry until the stalks are hard enough to snap when bent, they are then sent through a hemp break to break down the stalks, scutching, hackling and combing then follows once the stalks are broken down to separate the fiber and hemp hurds (figure 29)[74][75]. Recent developments have led to a machine called the decorticator which strips the skin, bark, leaves and seeds of the stalk. The development of this machine over the years has made hemp a more viable product to produce due to low labour costs but it places a negative effect on the environment. Using traditional methods places no damage on the environment besides for the retting process which contaminates the water, but bacteria within the water destroys the organic plant contaminants. Nonetheless the advancement in technology and development of the decorticator allows for plants to be harvested and sent through the machine, to end with separated hurds and fiber with no in-between process needed and large quantities being produced compared to the time consuming traditional methods by hand[76][77]. The time is now, new laws and technology allow for the reintroduction of hemp into the building sector (figure 30)[78][79].

Figure.....29



Figure.....30



New mind-sets and technology now allow for hemp to be used in construction. Used in the past but placed in the dark, a new light has uncovered its endless possibilities. These new laws have opened several doorways for the plant, the key to redirecting the human-made damage placed on Earth, *hemp*, a plant as a building material.

[70] (EMCDDA, 2018)
[71] (Herer, 2010)
[72] (EMCDDA, 2018)
[73] (CRS, 2019)
[74] (Boyce, 1900)
[75] (Lowery, A Jr. 2019)
[76] (Lyster H Dewey, 1916)
[77] (Lowery, A Jr. 2019)

[78] (Boyce, 1900)
[79] (Lowery, A Jr. 2019)

Hemp: Cannabis Sativa L, can be processed into a flower-based concrete called Hempcrete. A bio-composite building application with less than zero damage to the environment created by wet-mixing hemp shiv and a binder such as lime- forming a non-load bearing material cast into a mould (figure 37)[80]. The plant can grow consistently every year with no harm to the environment with a temperature range between 15-25 ˚C, and soil temperature 5.5c to 7.5 ˚C. The production of hemp is carbon negative, meaning it absorbs more carbon during its growth than it emits, an annual crop that can be regrown consistently to produce a building material. According to (Tradical, 2019), 1 hectare can produce 10 tons of hemp, which produces 6 tons of hemp shiv used to make hempcrete. (Nelson, 2000) explains that 118 sqm house can be produced from one hectare of hemp grown for fibre. Hemp shiv is made from processed hemp; the material itself is natural and can be regrown annually, a crop that can be regrown consistently to produce a building material (figure 31, 32,33,34) (figure 35).

The plant produces stalks refined to make hempcrete[81]. The cell structure seen in diagram B shows the different layers with the middle and bottom half of the plant producing the better parts for hemp hurds due to the thicker Xylem.[8283]. Hemp shiv or hemp-hurd is made from processed hemp stalks; the Xylem is extracted and separated from the rest of the plant; this core like structure is chopped down into pieces of hemp-hurds. Figure 35 shows the inner structure of the hemp stalk with the lower half, producing the best results for making hemp hurds. The ratio of hurd to lime binder to water varies depending on the mix; there is no set ratio yet, " 3:1:1 is a simple volume mix ratio," according to(Windsor, 2019). Lime is a typical binder that is used when creating hempcrete, but Tradical binders are common, used by UK Hempcrete as per (Windsor, 2019), figures 42 to 45 show the performance of Tradical hempcrete. The different types of lime binders are as follows: air limes, hydraulic limes, pozzolan, and formulated limes. Together with water and hemp shiv, they create the building application- hempcrete [84].

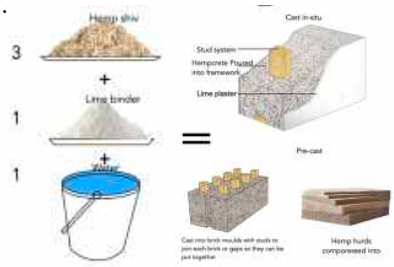


Figure.....36

Hempcrete is a multifaceted building application that can be used in various ways, but due to its non-load bearing flaw, it has to be cast into the structural system, framework and shuttering (figures 38 to 40). Once processed, it can be used in two different ways: cast-in-situ and pre-cast hempcrete, (figure 37). Cast-in-situ refers to mixing the hempcrete on site. Creating walls, floors, or roofing, the limitation with hempcrete is that it needs to be cast around a structural frame due to it being a non-load bearing material (figure 37). Ways of casting such as hand placing or spraying are most common when creating cast-in-situ hempcrete; pre-cast hempcrete refers to an alternative in-situ casting where hempcrete is pre-cast into blocks or framed panels. Pre-cast hempcrete comes dried and does not take time to set, allowing jobs to be completed with pace and ease; cast-in-situ takes several weeks to dry before finishing can be applied(Windsor, 2019). When cast around a structural system, hempcrete forms a material system [85](figure 39 & 40).

[80] (Stanwix & Sparrow, 2014)
[81] (Cole & Zurbo, 2008)
[82] (Stanwix & Sparrow, 2014)
[83] (Cole & Zurbo, 2008)
[84] (Stanwix & Sparrow, 2014)
[85] (Stanwix & Sparrow, 2014)

A plant, when harvested and refined, can be used to create a material building system known as hempcrete [86][87][88][89]. Its performance benefits are as follows: fire resistance, hygroscopic quality, resistance to moisture, excellent thermal mass, acoustic performance, and improved air quality. A natural material to provide a high level of thermal and acoustic performance, as well as a breathable wall that regulates moisture. The performance of hempcrete leads to better-performing buildings. It works by creating a "breathable wall" (Stanwix & Sparrow, 2014)(figure 46); which can be used for floor slabs (figure 41), walls (figure 39 &40) and roofing(figure 38) . The cellulose membrane allows for the slow release of heat or cold air stored in the building, leading to fewer fluctuations between the extremes of the temperature inside compared to the outside. (figure 46). To avoid cold bridges, the insulation at the plinth and foot of the wall are essential to ensure airtightness at the intersecting joints, a vapour-permeable plaster finish or render finish is also vital to ensure complete airtightness all round, leading to the wall naturally ventilating the building. The hygroscopic nature helps manage humidity levels which eradicate mould build-up, a major cause of sick building syndrome. The remaining two qualities are its resistance to fire as it forms a barrier between the timber structure and fire; the acoustic performance is similar to the thermal performance as the wall creates a buffer and absorbs sound (figures 47 to 49)[90].

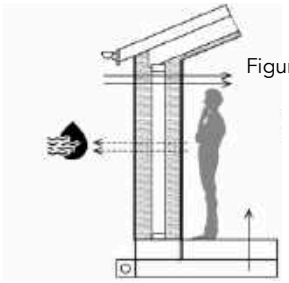


Figure.....47

Humidity control

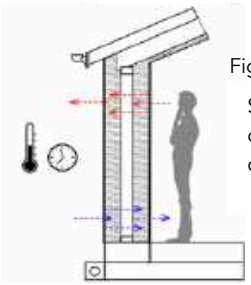


Figure.....48

Slow release of hot and cold air

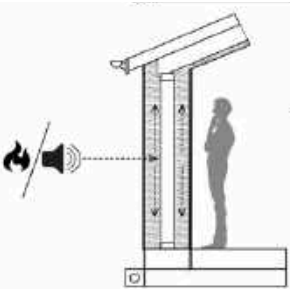


Figure.....49

Fire and sound absorption

Hempcrete can be used as a non-load bearing material system with the aid of structural elements such as timber to give it support. Forming a continuous mould around the framework, fewer materials are needed to complete the system, with only a render or plaster coat needed to finish the seal. Multi-faceted, it can be cast into any moulded shape, internally it can be used as an acoustic and thermal buffer between rooms and allows one to avoid using embodied carbon materials such as gypsum, concrete, and synthetic insulation. Its foundation uses are weak and can only be used as a floor slab, as it is vapour permeable it cannot be drenched or exposed to water for long periods as it will result in deterioration or failure; a plinth is needed to protect the floor slab from water splash back. (figure 41). The structural frame is placed onto the plinth or foundation, using studwork such as floor plates, studs, wall plates, to form the frame, the frame needs to be alkali resistant to prevent corrosion; usually enclosed centrally within the wall but can be positioned flush with the surface of hempcrete internally or externally (figures 39 to 41). The shuttering is important to hold the hempcrete while it dries; it can either be permanent or temporary (figure 39). The framework and shuttering form the structural frame, and any openings for doors and windows are created as part of this frame. The typical thickness of a wall is around 300-400mm to achieve a U-value of 0.2-0.15 W/m2k, which changes with size, a standard wall is between 300-350mm [91] As a material system, it behaves differently to most, with many performance benefits. It is a plant with varied uses, a seed to birth new materials for enhancing life.

[86] (Boyce, 1900)
[87] (Herer, 2010)
[88] (Stanwix & Sparrow, 2014)
[89] (Nelson, 2000)
[90] (Stanwix & Sparrow, 2014)
[91] (Stanwix & Sparrow, 2014)

This building proves that hemp can be used as a material system to create a low-embodied renewable building. The center for alternative technology (CAT) in Wales, constructed a "State of the art educational facility" made from Hemp. The use of hempcrete is in the makeup of all the walls [9293]. This building is evidence of how hemp, a plant, can be used as a material system to create a contemporary sustainable building. The building was completed in 2010 by David Lea and Pat Borer Architects, winning a RIBA award in May 2011[94]. Designed to have a low environmental impact in construction and use, CAT is made from: FSC (Forest Stewardship Council) certified whitewood glue-laminated timber, hempcrete, rammed earth, lime renders, cork and sand-lime bricks for the foundation [83]. Using modern methods to create a low-energy demand building to reduce the environmental impact both in the make-up and function of the building [9596]. Spaces maximize daylight and sunlight to make people aware of time and the weather, using windows and louvres. Rooms are surrounded by courtyards giving a feeling of space and connection with the outside world, users can open the area to the sounds and smells of nature (figures 50 to 52)[97]. Shown in the plan, the walls lead users to their desired rooms with ease and are thicker in diameter to create the desired U value. The main room, the Sheppard theatre which uses top light to light the space, and can be opened or closed depending on the conditions of the day as seen in (figure 56)[98]. The walls and theatre show the performance of hempcrete and how it can be used to create large curved open spaces. (figure 55&57).

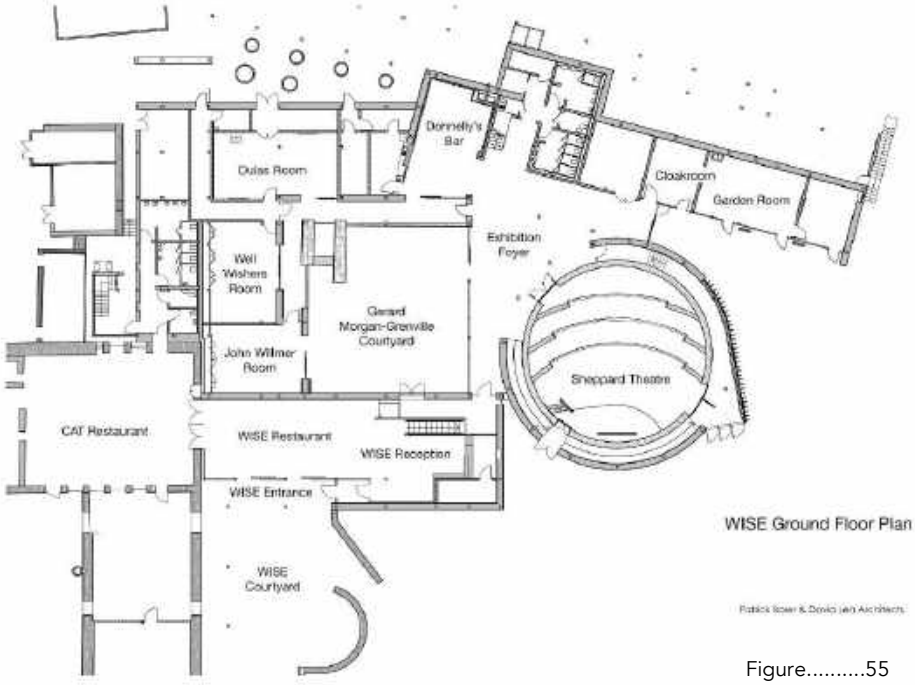


Figure.....55

[92] (EUmiesAward., 2019)
[93] (CAT, 2019)
[94] (EUmiesAward., 2019)
[95] (EUmiesAward., 2019)
[96] (CAT, 2019)
[97] (EUmiesAward., 2019)
[98] (EUmiesAward., 2019)

Strategies are used to enhance the performance of the building. Natural ventilation is used throughout; passive stack ventilation is created by positioning vents on either side of the room and at different levels, both high and low to create an up draught of warm air [99]. Windows and louvres are added to maintain natural ventilation and prevent overheating in summer--including casement vents, timber shutters, and fly screens which improve the natural ventilation [100]; all aimed to reduce heating, ventilation and air-conditioning (HVAC) costs.[101](figure 58). In-order to lift the hempcrete walls off the floor, a raft foundation is used to ensure continuous water flow is kept away. Water collection is also used to redirect rain from the walls into gutters; runoff cascades from the terraces, gutters are positioned into the central courtyard which drains water into the old quarry's waterwheel pit, as seen in the sketch below (Figures 53&54) [102][103]. The walls, strategies and foundation show the performance of the building and how it can create large curved spaces, using a plant-based material system, hempcrete.

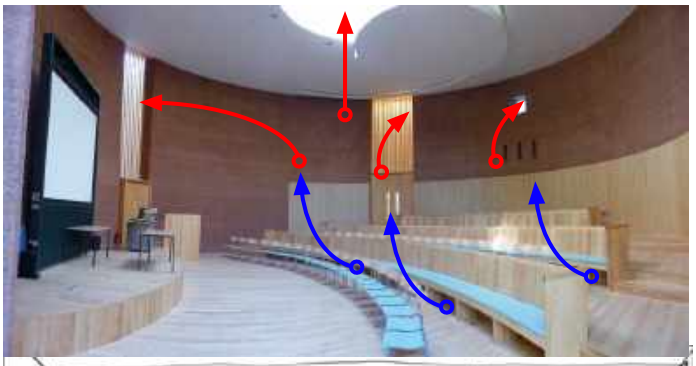


Figure.....58

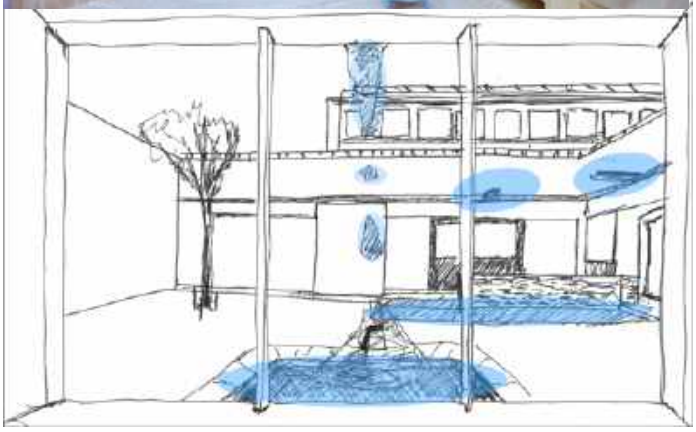


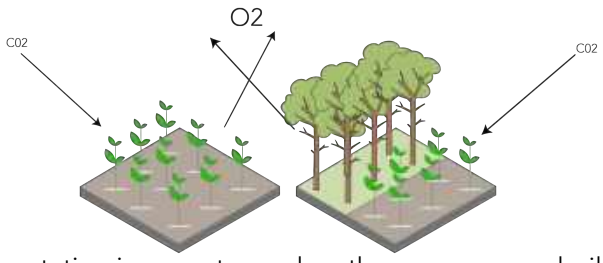
Figure.....54

The building is an example of how hemp is used to produce a low-embodied energy building. The space ranges in size and shape, providing various functions for the users, a building made from a plant that varies in form and function. A low-embodied piece of architecture that has very little impact on the environment. Evidence to show how hemp, when combined with a structural system, produces a sustainable low impact building bringing environmental benefits to the building sector.

[99] (EUmiesAward., 2019)
[100] (EUmiesAward., 2019)
[101] (EUmiesAward., 2019)
[102] (CAT, 2019)
[103] (EUmiesAward., 2019)

Rising environmental concerns call for solutions. "Changing the world requires unconventional responses (Leonardi & Stagi, 2019).The Green Economy report highlights some of the solutions, steps to new greener pathways, aimed to prevent the loss of biodiversity and ecosystem services, describing a green economy as improved human well-being and social equity, while significantly reducing environmental risks and ecological sacrifices. The report put forward by UNEP is a result of growing climate problems [104]. Shown in the IPCC report on global greenhouse gas emissions (GHG), which highlights global environmental concerns and what happens when global warming increases above pre-industrial levels from 1.5c to 2c. Impacts on: biodiversity with species loss and extinction due to the inability to cope with the changing environment, a change in temperature, and thawing permafrost leading to fires, invasive species, coral reef loss. The direct risks to human beings are health, livelihoods, food security, water supply, human security, and economic growth [105] — the main contributors being the transport and building sector as stated by (IPCC, 2018). The IPCC report put forward several sustainable goals such as lower transport and building demand, low-carbon energy use, and increasing afforestation as well as reforestation [106]. These reports form the discussion of how timber and hemp can be used to bring positive environmental and socio-economic benefits to the global population and will be explored further. Figure 59, a graph which shows how energy consumption in the building sector has the highest chance of mitigation. It shows the carbon prices placed on the building sector as well as the potential mitigation of 5.3-6.7 GT co2 yr (UNEP, 2011).

Figure…60:Carbon



Using reforestation and afforestation is a way to use less than zero-energy building materials which are vital to exploring the mitigation pathway as plants act as carbon sinks, due to carbon being absorbed during a plant's growth cycle. "Construction of new green buildings… can achieve significant savings." (UNEP, 2011). Hemp, "During the growth period "can sequest 15 tons of CO2 per hectare" stated by (Wilson, 2016, p. 2), further backed by (Tradical, 2019)) which explain that one hectare can sequest "18 tons of CO2". When processed, it picks up 10 to 11 tons leaving 4 to 7 tons being absorbed. Specific to Tradical Hempcrete, the wall mix contains 150 kg of CO2/m3, and the roof mix contains 155kgm3. Depending on the size of the wall and house: a 52 sqm with a 300mm wall thickness sequest's up to 5.4 tons, and a 100 sqm house with a 500mm wall thickness sequest's up to 13.2 tons, per (Tradical, 2019). As no pesticides or herbicides are used, it is environmentally safe; it is rated top 5 out of 25 crops when looking at protecting biodiversity, and it can restore soil health through phytoremediation (figure 64)[107]. Trees, when harvested in responsibly managed forests, can lead to a reduction in carbon emissions similar to hemp, as they both have benefits arising from carbon sequestration during the growth period, with forest carbon sinks in the UK reducing 4t C ha year [108] Using plants to create whole buildings will decrease the amount of carbon dioxide produced by the building sector. A resource to perform as a combined structural and material system that is renewable and sustainable, complex in form and function to solve modern-day issues that are threating the survival of all species on planet earth.

[104] (UNEP, 2011)
[105] (IPCC, 2018)
[106] (IPCC, 2018)
[107] (Wilson, 2016)
[108] (Michael H. Ramage, 2016)

Plants require care, which means a large labour force is needed. The socio-economic benefits lie in the number of jobs and social wealth that is produced when plants are used in construction. The processing line within hemp and timber is extensive [109]. There is a need for a large workforce to ensure that production does not drop [110]. Stated by (Johnson, 2018, p. 1) hemp produces around 25 000 products with "U.S retail sales $700 million" (Johnson, 2018, p. 1)(Figure 62). It may replace existing industries, but its wide variety of uses means a wide range of job creation. It is labour-intensive and requires many workers from start to finish due to the extended supply chain (figure 61). In 2016 the European market produced hemp on 80 000 acres; a large amount of space needs a large workforce to cope with the up keep [111]. Industrial Roundwood used to create timber for construction is valued at $103.4 billion, an example of how timber is highly valued, a material to produce jobs to maintain the much-needed supply (Figure 63) [112]. It can be sustainably harvested without depleting or degrading forests, as seen in the WISE building, which uses FSC certified whitewood glue-laminated timber [113]. Managing such forests to cope with the extremely high value means a workforce is needed to produce the products. Production to processing, plants lead to job creation, a renewable and sustainable building material, producing jobs to stimulate economic growth while saving the planet.



Figure…65: Trees produce jobs leading inflow of money

The growth and production of plants create healthier environments for all living things, which increases social wealth due to improved air quality from the growth cycle of plants. Social-welfare due to job creation and improved environmental benefits is the result of the market size when used by the entire building sector; a renewable, sustainable economy can be created. A solution for today, tomorrow, and forever on. Plants as a structural and material system to keep the lights on.

[109] (Michael H. Ramage, 2016)
[110] (UNEP, 2011)
[111] (Johnson, 2018)
[112] (Michael H. Ramage, 2016)
[113] (CAT, 2019)

Concluding, it can now be said that plants as a structural and material system; reduce carbon dioxide emissions and provide socio-economic change. A plant, a flower, a tool to redirect the course of humanity can be used as a structural and material system. Nature is complex and diverse, formed before our time; it produces life to all beings that live on planet earth. A delicate resource that was created to last forever, now facing extinction, new radical ways of thinking and thus designing are needed. Art nouveau revolted against what was done before as the pioneers understood the faults in the architecture of the time. This resulted in an organic form of architecture, using tools to reconstruct the world. Frei Otto descended from this way of thinking. Understanding that humans need to work with nature rather than opposed to nature.

"Biggest enemy of nature is man" (Otto, 2003)

Nature is not our enemy; we are nature; nature is us. Biometric design was used to form his structural system, mimicking the forces found in plants he created a design that minimised material waste and maximized force performance — using nature to inform design, with minimal damage. Using materials that consume carbon in their production leads to a less than zero-energy building material. Hemp in its growth reduces environmental damage and produces a renewable, sustainable building resource, when refined it turns into a resource that can be turned into a material system, with its non-loading bearing flaw everted through the structural system.

"Ecological approach to reduce…construction energy" (John Fraser, 1995, p. 18)

Using nature to inspire and form a building system will reduce the building sectors energy demand. Absorbing carbon dioxide and providing jobs, plants through an ecological approach to design will revolutionize the building sector, not only for the buildings themselves but for all those who live on planet earth. Further on, a design thesis project will be formed to examine if the structural system and the material system can be combined to form a less-than-zero energy building through enhanced performance.

Informing my design thesis project

A structural system was developed concluding the research drawn from Frei Otto's structural system. In my project, I aimed to direct water for water collection and support the material system with minimized material usage. A study was drawn around hempcrete to test and experiment with the material to assess how it works so I could further develop my design. Developing the structural system and material system alongside one another to assess weather a plant can produce both systems and produce a better performing building



Hanging model test

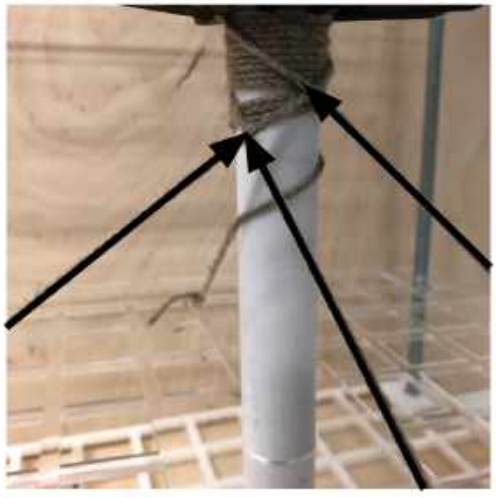
I constructed a hanging model to assess how to structure the support system to carry water between levels. I ended with branches using compression to disperse force



Tension

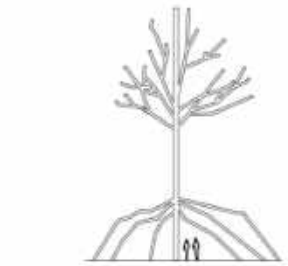


Testing ways to use branching aimed to find the best way to direct water with maximum structural performance

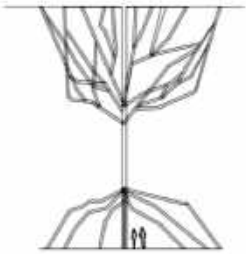


Tension

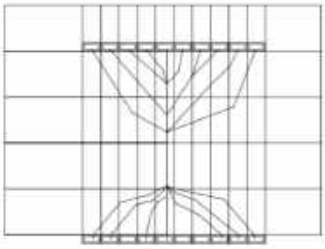
Development of the structural system to collect water and support loads



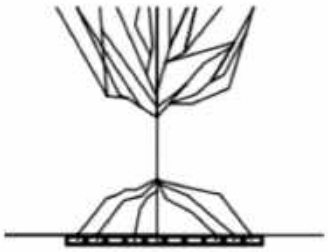
Branching of a tree



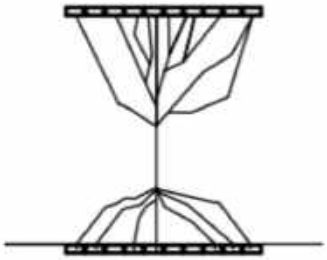
Branches moved to support a roof or floor



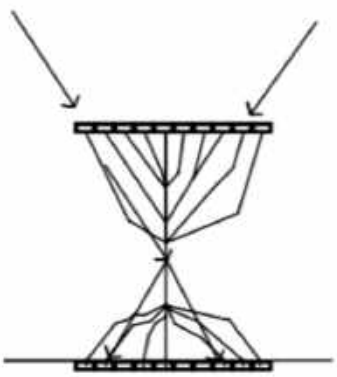
Grid setup and used to move branches for rain collection



Water is collected at the top and passed into the grid



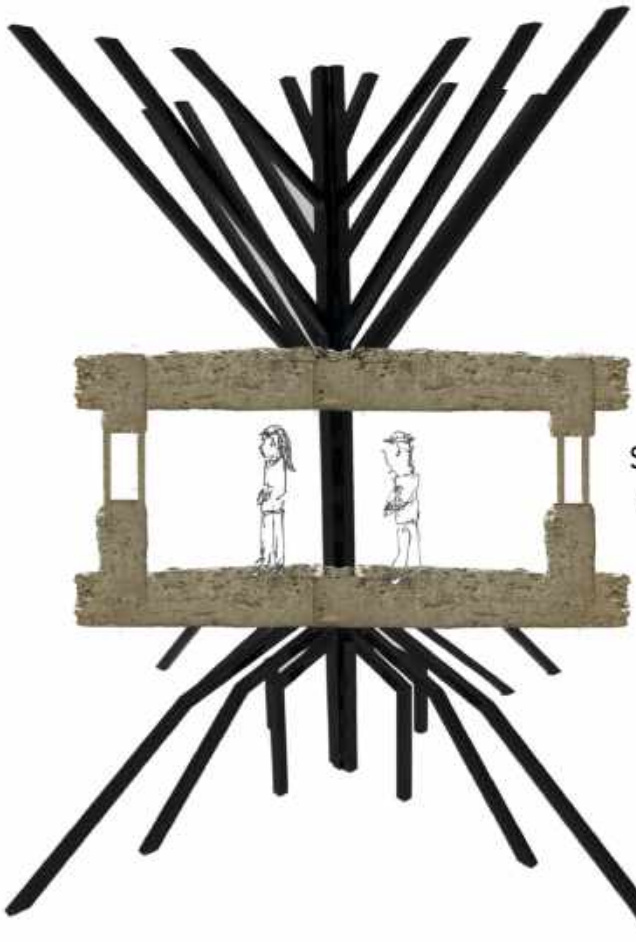
Water is transfered through the branches to the water collection within the grid

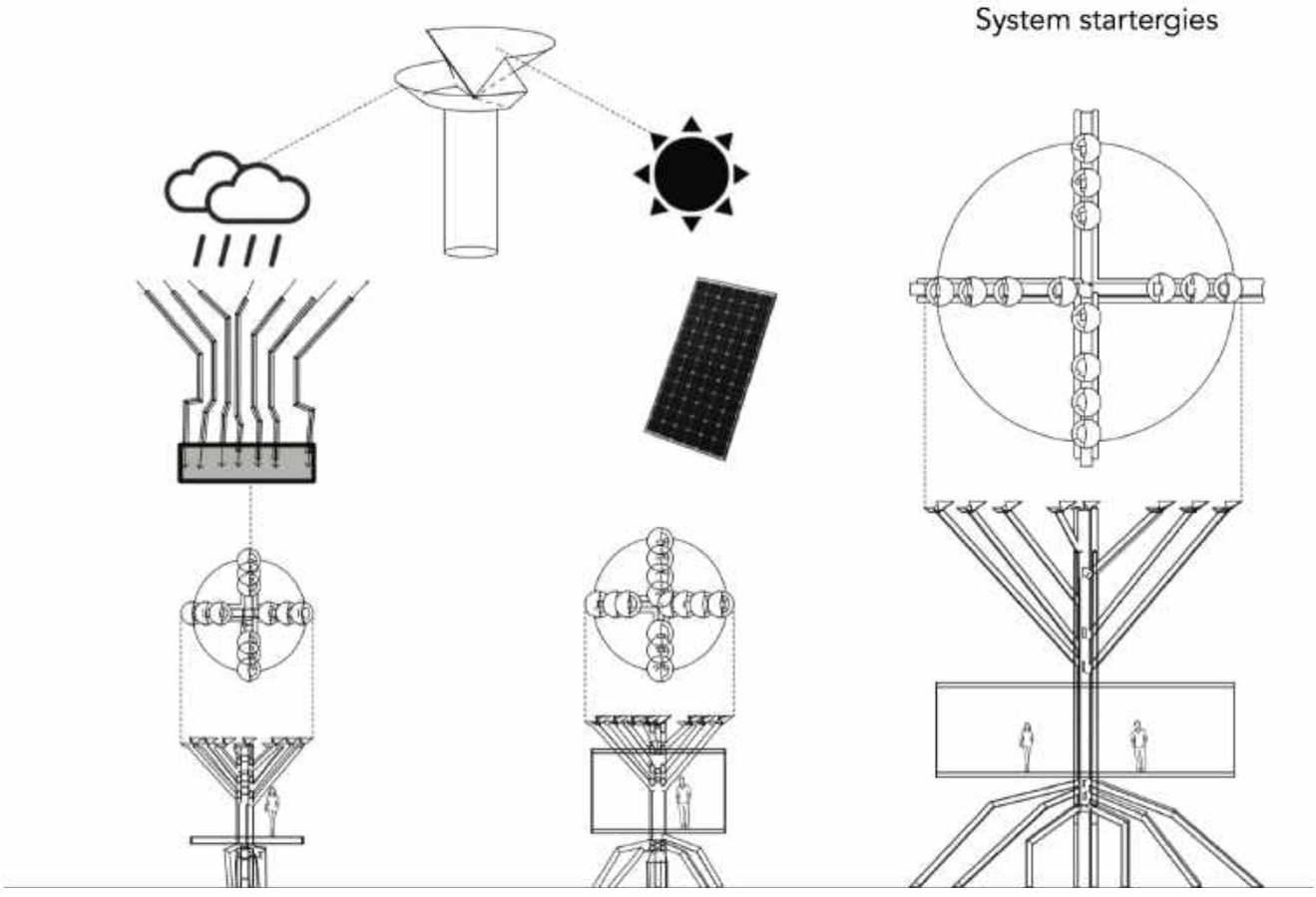


Waters flow through the structure

Outcome

This is the start of a concept, mixing the structural system and material system to form one whole building. The intended strategies are rain water and solar collection with minimized material usage.





Appendix's

Appendix .....A: Nature as an inspirational paradigm
Appendix .....B: A plant as structural system
Appendix .....C: Plant as Material system
Appendix .....D: Environmental benefits and Socio economic benefits

Appendix .....A: Nature as an inspirational paradigm

Figure list .....1: Victor Horta. (2019). Retrieved 20 September 2019, from https://en.wikipedia.org/wiki/Victor_Horta

Figure list .....2:Major Town Houses of the Architect Victor Horta (Brussels). (2019). Retrieved 20 September 2019, from https://whc.unesco.org/en/list/1005/gallery/&index=13&maxrows=12

Figure list .....3: Antoni Gaudí. (2019). Retrieved 20 September 2019, from https://en.wikipedia.org/wiki/Antoni_Gaud%C3%AD

Figure list .....4: Major Town Houses of the Architect Victor Horta (Brussels). (2019). Retrieved 20 September 2019, from https://whc.unesco.org/en/list/1005/gallery/&index=13&maxrows=12

Figure list .....5: Antoni Gaudí. (2019). Retrieved 20 September 2019, from https://en.wikipedia.org/wiki/Antoni_Gaud%C3%AD

Figure list .....6: Estos son los 47 tesoros del patrimonio de España con el sello de la Unesco. (2019). Retrieved 20 September 2019, from https://theworldnews.net/es-news/estos-son-los-47-tesoros-del-patrimonio-de-espana-con-el-sello-de-la-unesco

Figure list .....7: Estos son los 47 tesoros del patrimonio de España con el sello de la Unesco. (2019). Retrieved 20 September 2019, from https://theworldnews.net/es-news/estos-son-los-47-tesoros-del-patrimonio-de-espana-con-el-sello-de-la-unesco

Appendix .....B: A plant as structural system

Figure list .....8: frei otto: a life of research, construction and inspiration. (2019). Retrieved 20 September 2019, from https://www.designboom.com/architecture/frei-otto-a-life-of-research-construction-and-inspiration-detail-book-report-06-02-2015/

Figure list .....9: frei otto: a life of research, construction and inspiration. (2019). Retrieved 20 September 2019, from https://www.designboom.com/architecture/frei-otto-a-life-of-research-construction-and-inspiration-detail-book-report-06-02-2015/

Figure list .....10: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list .....11: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list .....12: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list .....13: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list .....14: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list .....15: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list .....16: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list ……17: Nerdinger, W., Meissner, I., Möller, I., & Grdanjski, M. (2005). *Frei Otto: Complete Works* (2nd ed., pp. 10-250). Basel: Birkhäuser.

Figure list ……18: Nerdinger, W., Meissner, I., Möller, I., & Grdanjski, M. (2005). *Frei Otto: Complete Works* (2nd ed., pp. 10-250). Basel: Birkhäuser.

Figure list ……19: Nerdinger, W., Meissner, I., Möller, I., & Grdanjski, M. (2005). *Frei Otto: Complete Works* (2nd ed., pp. 10-250). Basel: Birkhäuser.

Figure list ……20: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list ……21: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list ……22: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Figure list ……23: Otto, F., & Rasch, B. (1996). *Frei Otto, Bodo Rausch* (2nd ed., pp. 10-230). Finding Form: Towards an Architecture of the Minimal: Stuttgart.

Appendix ……C: Plants as Material system

Figure list ……24: SAINT-CÉNERI-LE-GÉREI - The Baguette. (2019). Retrieved 20 September 2019, from http://baguette.over-blog.net/article-saint-ceneri-le-gerei-56759603.html

Figure list ……25: Campos, H. (2019). The Evolution of Hempcrete and Hemp Construction. Retrieved 17 October 2019, from https://www.thcint.com/thcint-blog/the-evolution-of-hempcrete-and-hemp-construction

Figure list ……26: Campos, H. (2019). The Evolution of Hempcrete and Hemp Construction. Retrieved 17 October 2019, from https://www.thcint.com/thcint-blog/the-evolution-of-hempcrete-and-hemp-construction

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Figure list ……33: Hemp, E. (2019). Why you should be using hemp | East Yorkshire Hemp. Retrieved 15 August 2019, from https://eastyorkshirehemp.co.uk/why-use-hemp.html

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Appendix .....D: Environmental benefits and Socio economic benefits

Figure list .....59: : UNEP, 2011, *Towards a Green Economy: Pathways to Sustainable Development and Poverty Eradication - A Synthesis for Policy Makers*, www.unep.org/greeneconomy

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Figure list .....65: personal diagram

Appendix……A: Nature as an inspirational paradigm



Figure……..1

Victor Horta

6 January 1861- 8 September 1947

A founders of Art Nouveau, one of the first architects to include electric lighting into building (Forshaw, 2016)



Figure……..2
Interior of *Horta Museum*

Using steel and timber to create decorative coral forms (Forshaw, 2016)

Appendix……1: Nature as an inspirational paradigm



Figure…..3
Paul Hankar -11 December 1859-17 January

He was a belgian architect and an innovator in art nouveau (Forshaw, 2016)



Figure…..4

Hotel Ciamberla

Geometric floral forms (Forshaw, 2016)

Appendix……A: Nature as an inspirational paradigm



Figure…..5

Antoni Gaudi

25 June 1852-10 June 1926

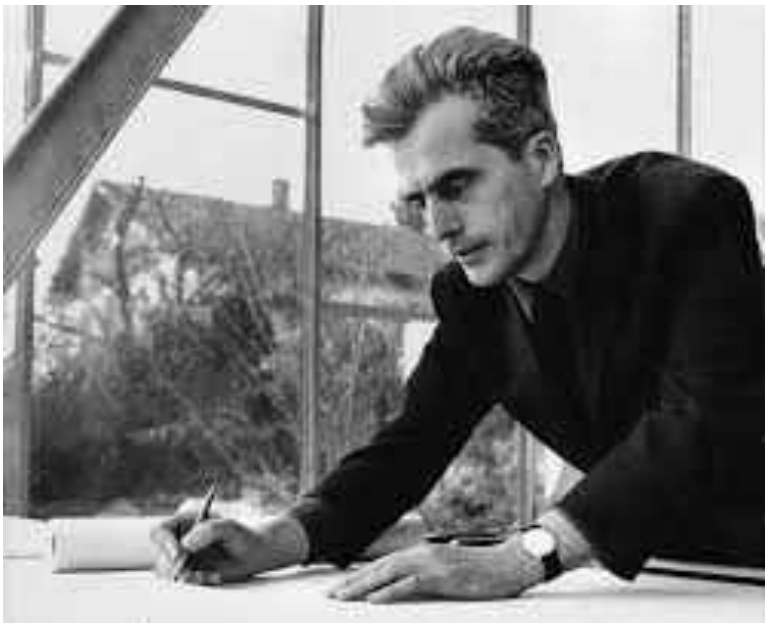


Figure…..6
Intricate design of Basilica de la Sagrada Familia



Figure…..7
Basilica de la Sagrada Familia

Appendix……B: Plants as structural system, Frei Otto



Figure…..8

Frei Otto

31 May 1925- 9 March 2015

A scientist, architect, and philosopher inspired by natural forms, a pioneer with responses to rising environmental concerns. (Nerdinger, 2005)

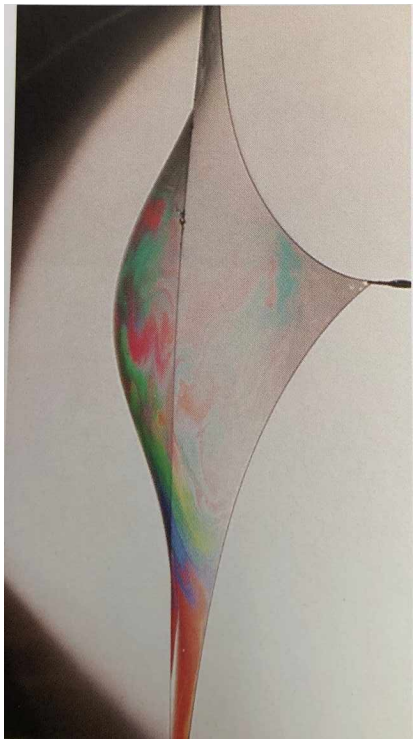


Figure…..9 Structural system inspired by the forces within plants and trees, to minimise material usage and maximise the load bearing force.

Appendix……B: Soap film: Frei Otto, Experiment's

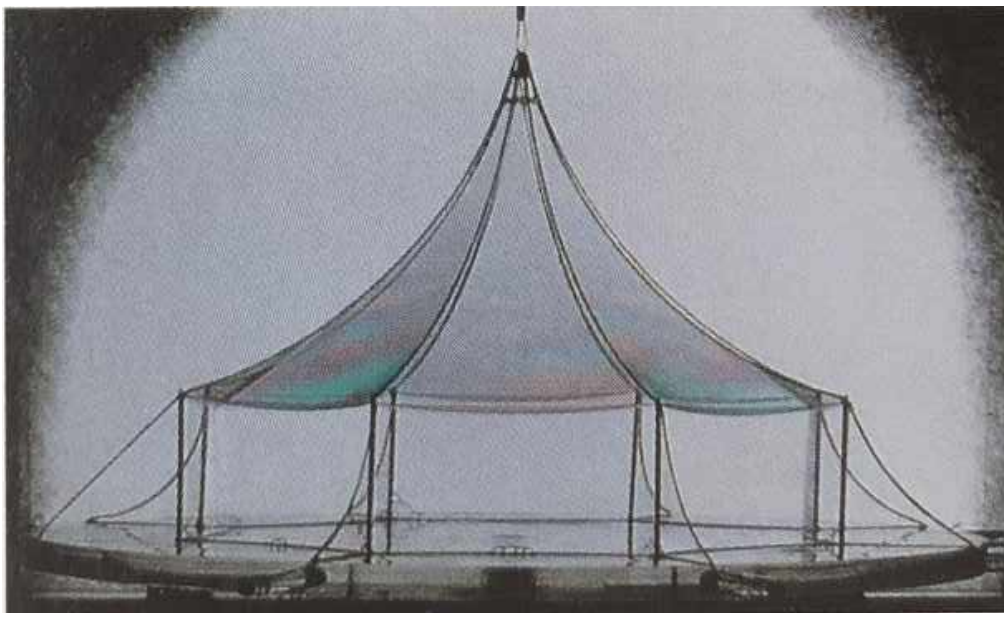


Figure……10
Soap film membranes to produce minimal areas as form finding models for tension loaded constructions



Figure……..11

Soap film in wind tunnel



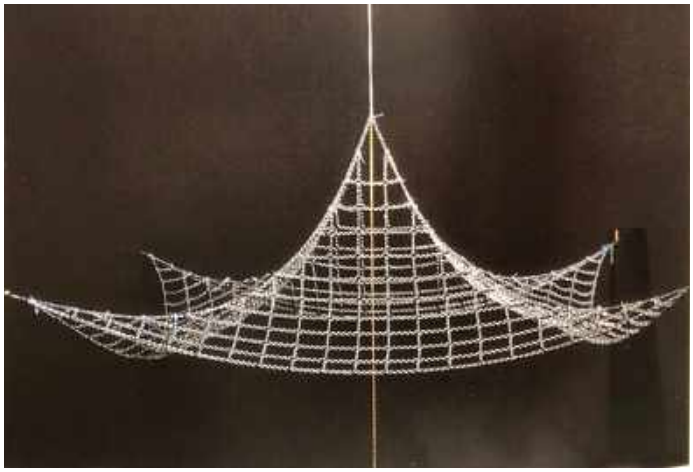
Figure…….12
Tent structure

Appendix……B: Chain tests Frei Otto, Experiment's



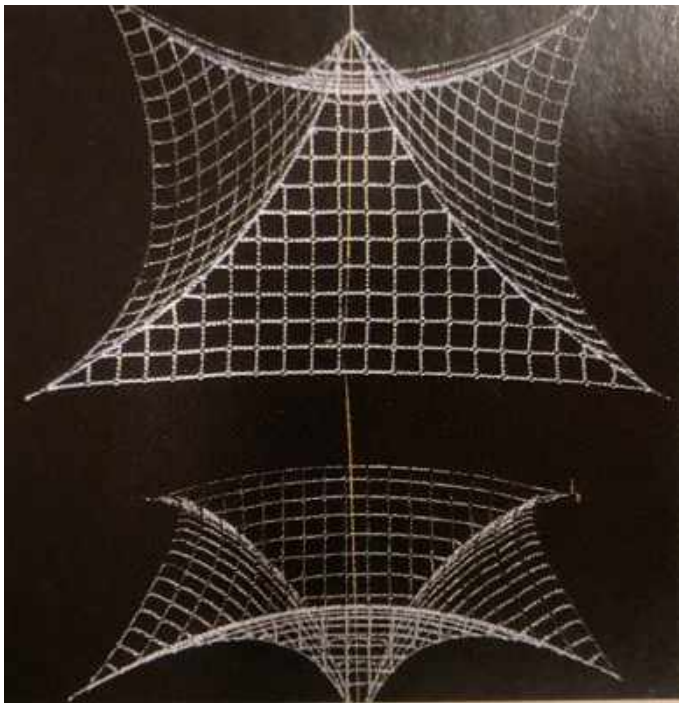
Figure.....13

Experiments with chains and nets



Figure……14

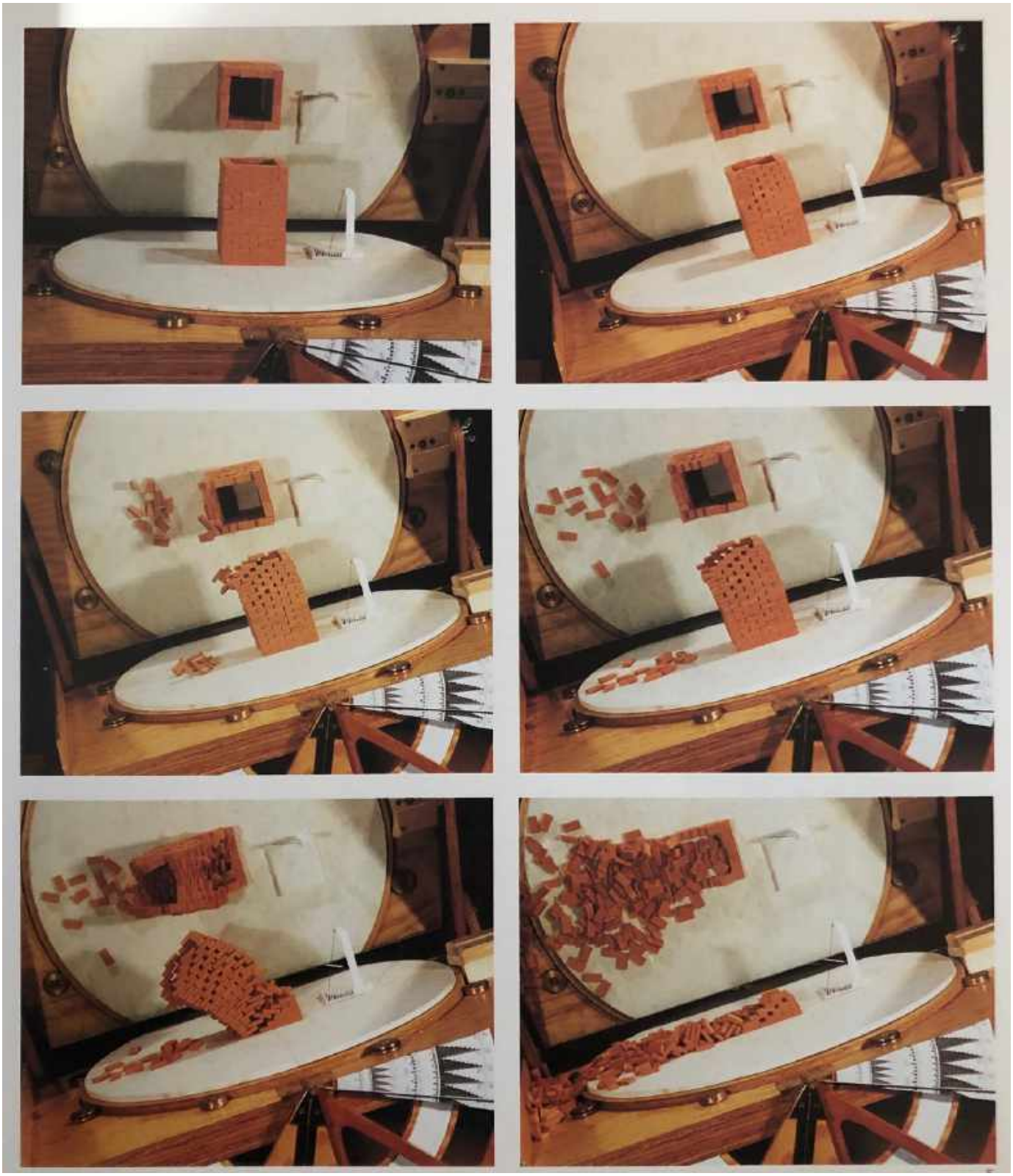
Suspended model using chain mesh
to form the shape of Asian roofs



Figure……15

Suspended models with triangular and square
mesh

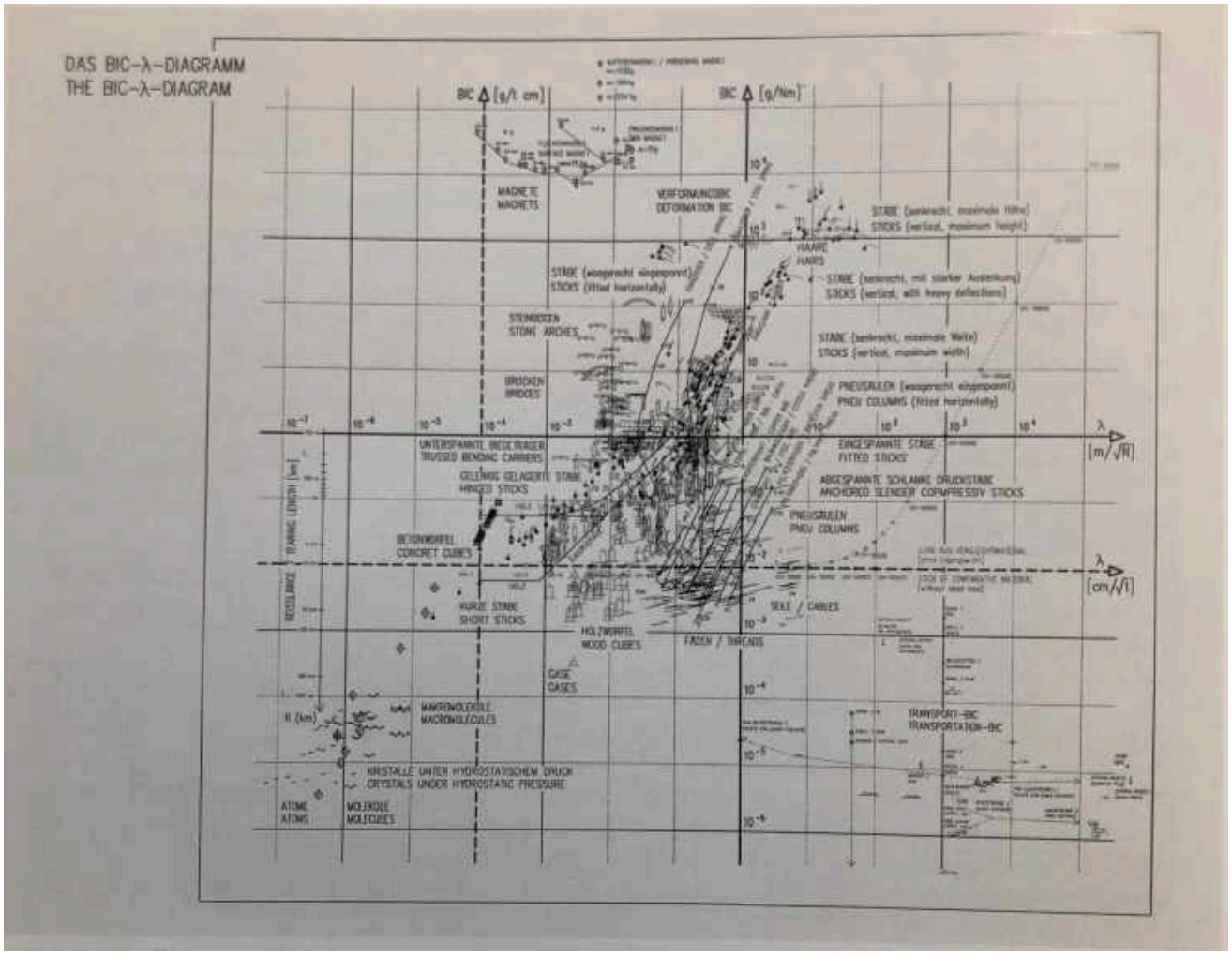
Appendix……B: Stability testing Frei Otto, Experiment's



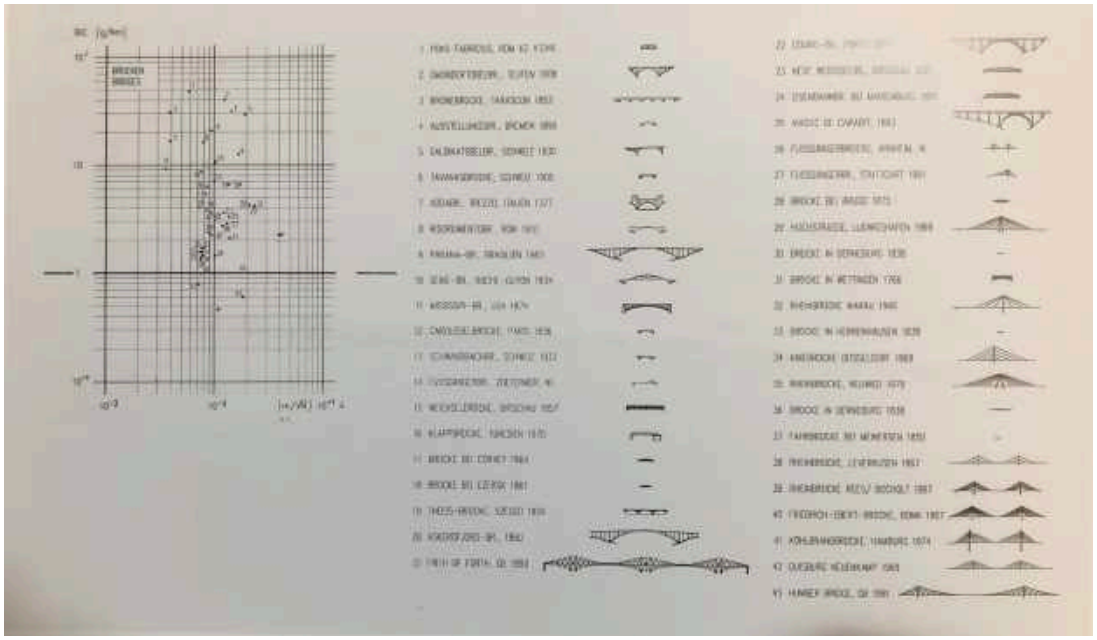
Figure……16

Testing the stability of buildings by placing a model on a tilting turntable then tilting and turning it to test the brick material. Used to test stability of constructions doing earthquakes

Appendix……B: B-I-C formula: Frei Otto, Experiment's



Figure……..17: BIC diagram to assess lightweight use of different structures
BIC = m /Tra. Tra was used to classify structural forms. Tr= Ti/Ta. Ti is the product of force and distance and Ta is the product of all external loads and the distance from the supports.
λ=s/√F is the relative slenderness formula



Figure……..18: BIC diagram used in bridge constructions

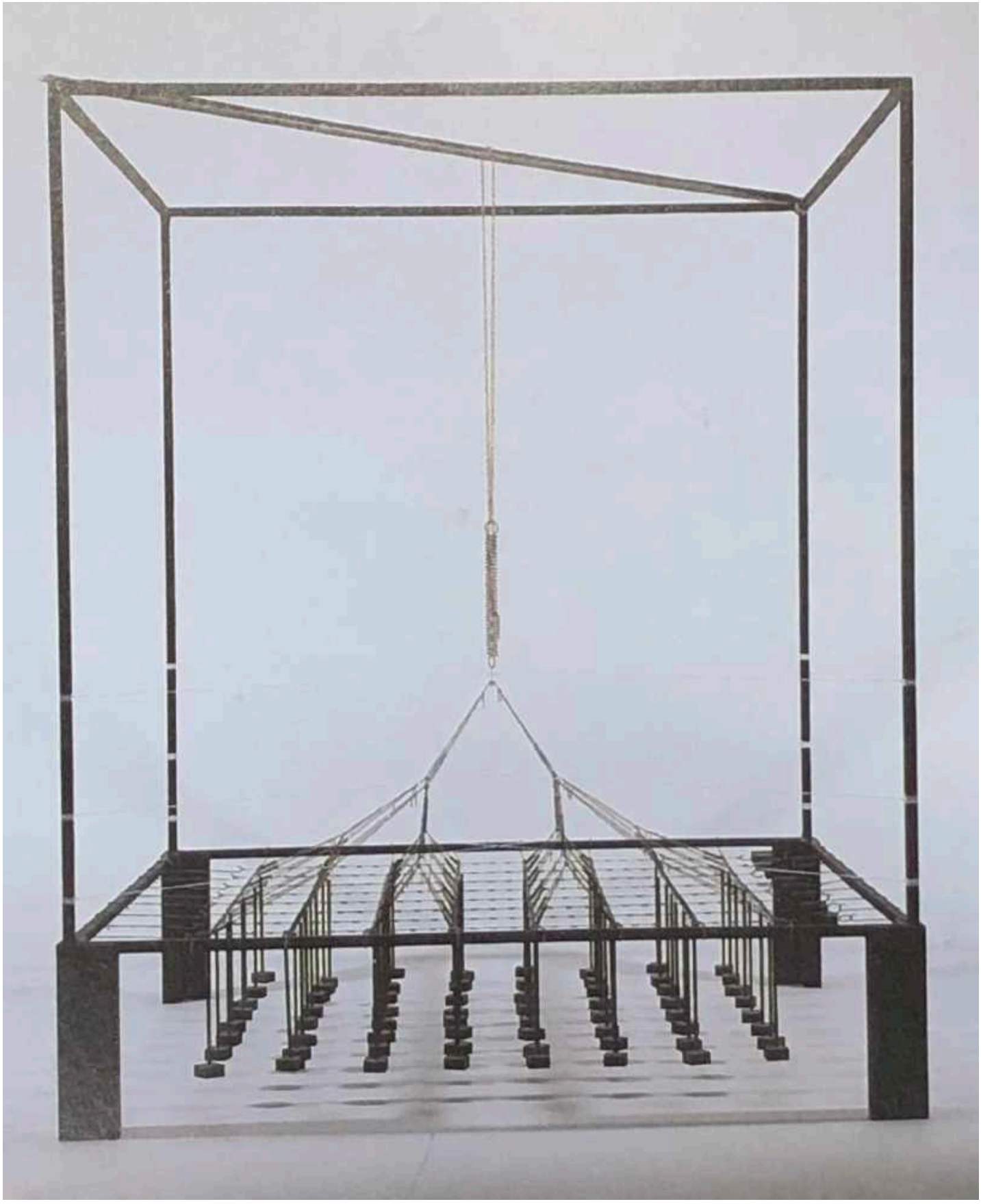
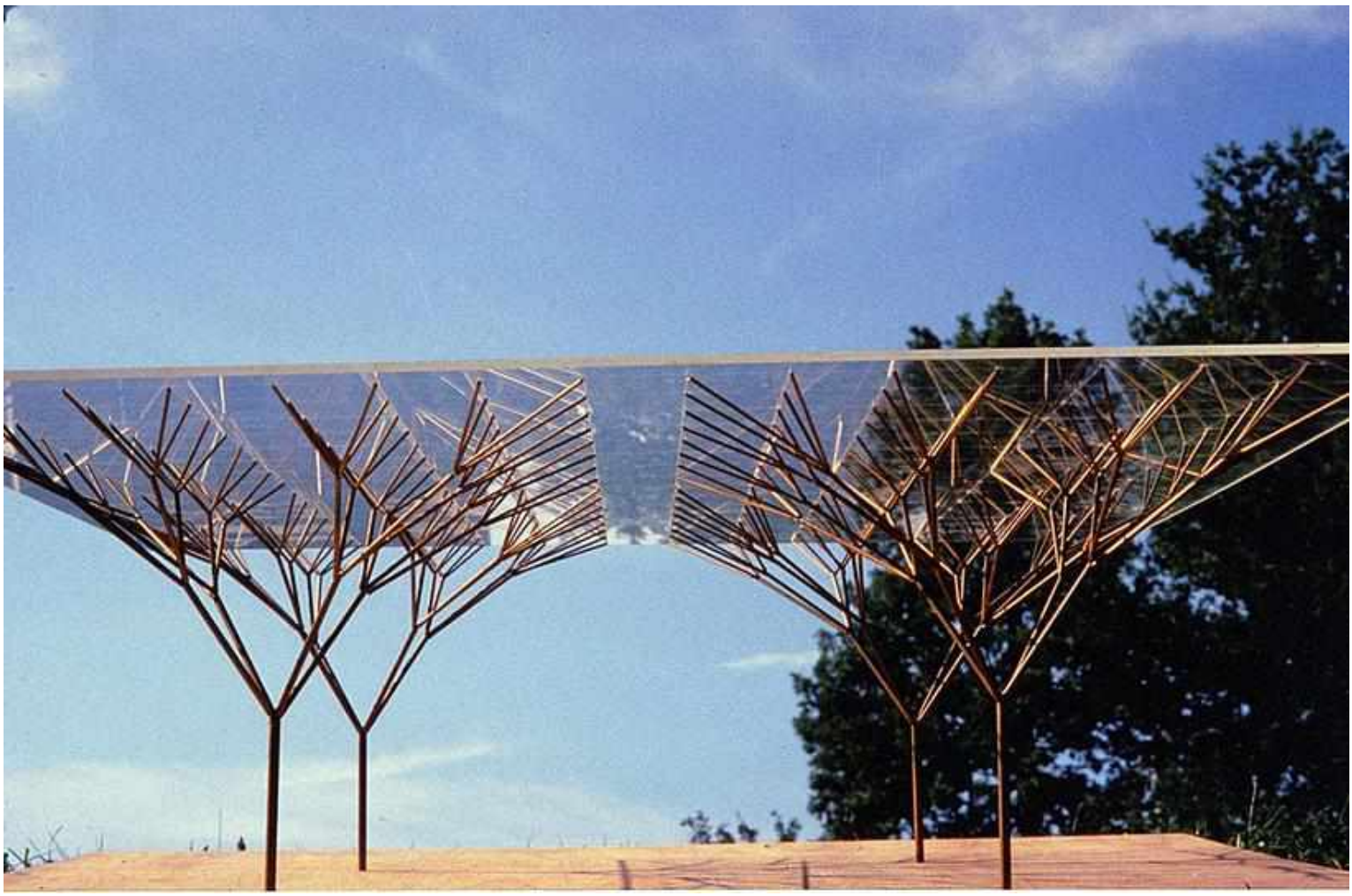


Figure.....19

Determining geometry from a suspended form

Appendix……B: Structural branching system, Frei Otto.



Figure……20
Model of the structural system

Figure.....22
Drawing of leaf structure



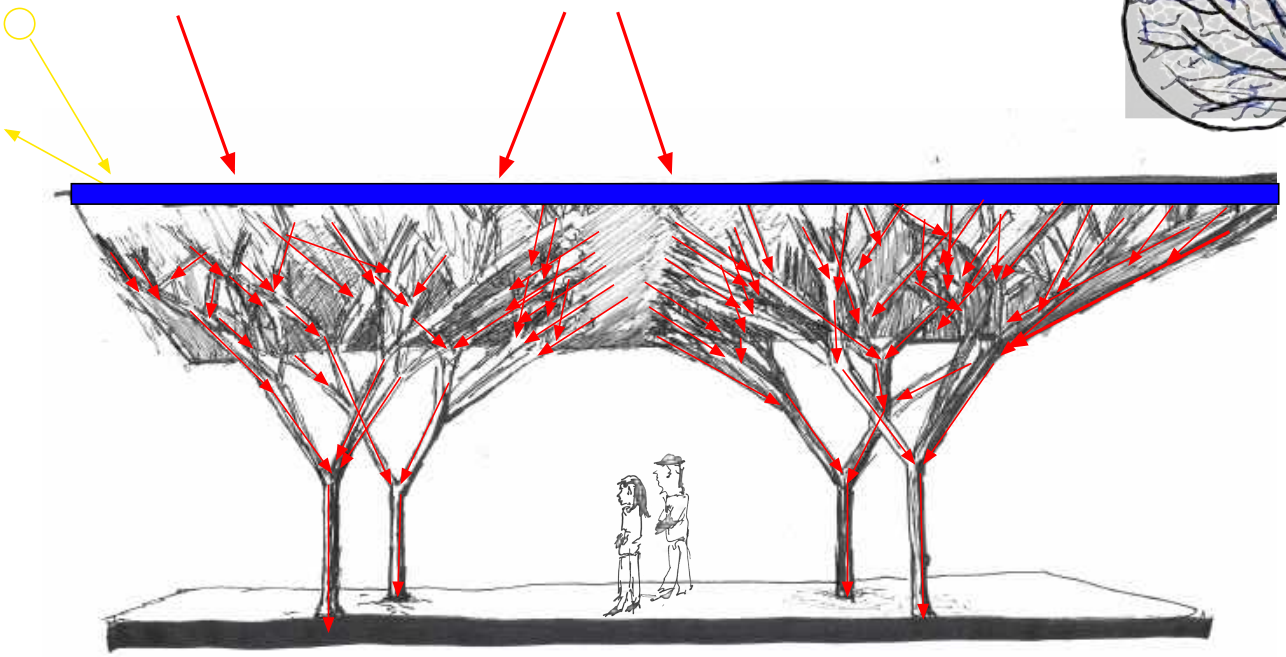


Figure.....21
Drawing of the structural system to show open space, shading and force being dispersed

Branched constructions as three-dimensional supporting systems

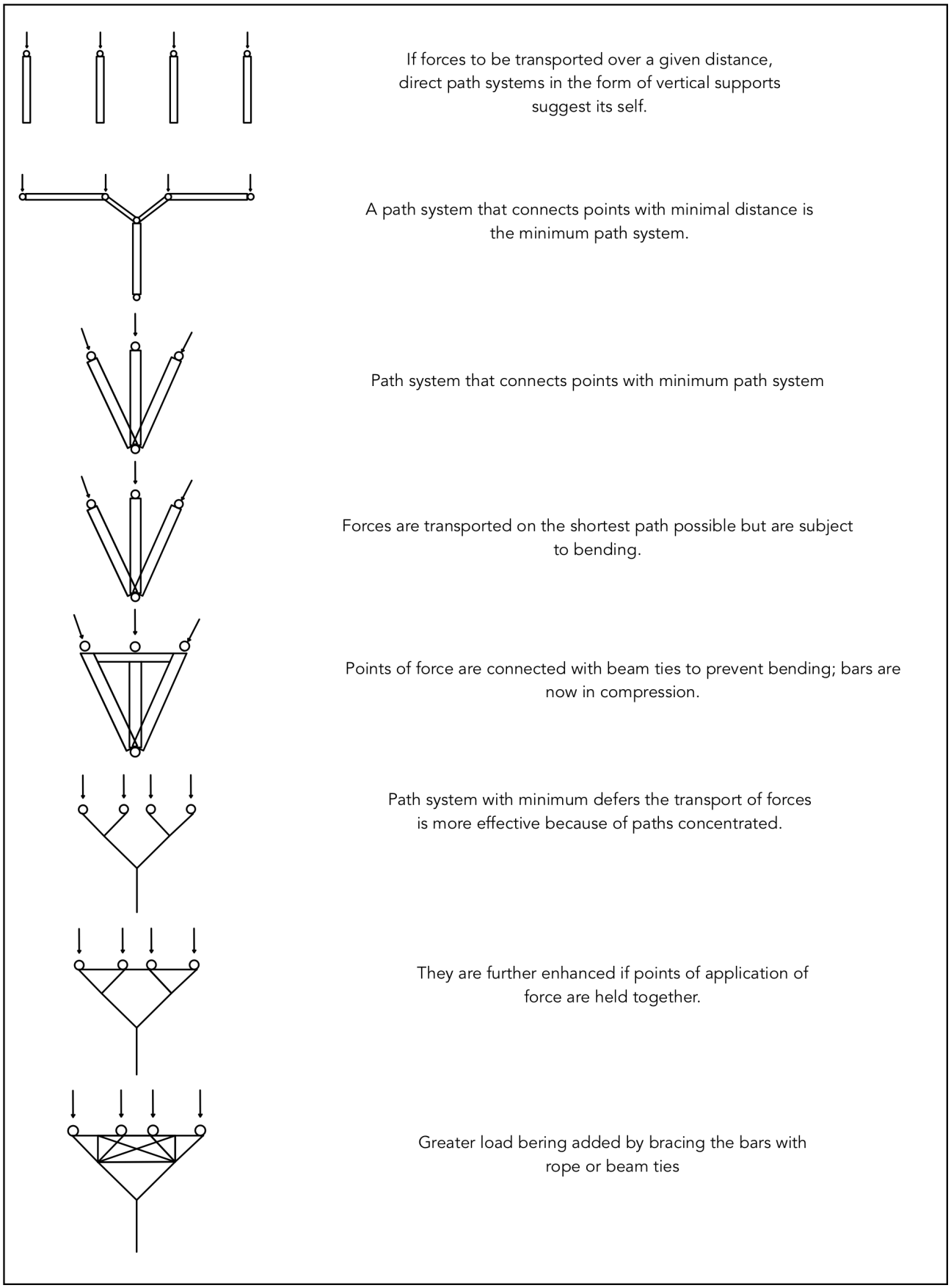
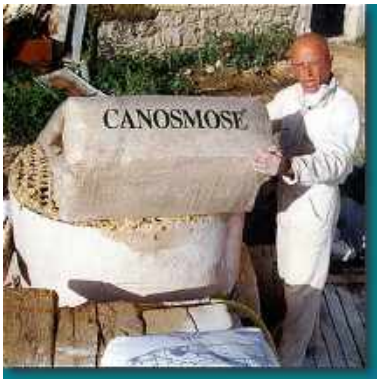


Figure.....23
Diagram showing development of structural system.



Figure……..24: Merovingian bridges, 600 AD
Hemp was used in mortar to construct the bridges. It shows the sustainability and durability of the renewable plant used in a material system



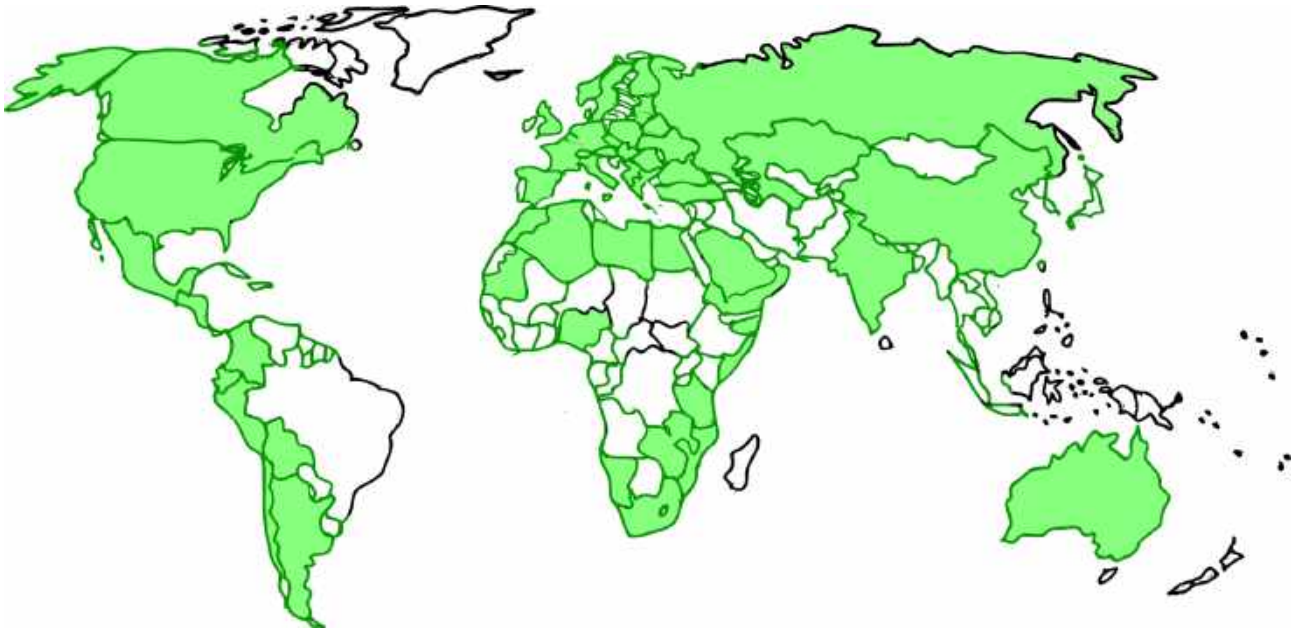
Figure……..25: Yves Kühn

The inventor of hempcrete



Figure……..26: Yves Kühn private house

The construction of the first house made from hempcrete



Figure……28: Countries that have legalised hemp

Argentina, Austria Belgium, Belize, Bosna i Hercegovina, Bulgaria, Canada, Chile, Colombia, Costa Rica, Croatia, Cyprus, Czech Republic, Estonia, Finland, France, Georgia, Germany, Greece, Guatemala, Hong Kong, Hungary, India, Ireland, Italy, Latvia, Lithuania, Luxembour, Malta, Netherlands, New Zealand, Northern Ireland, Norway, Paraguay, Peru, Poland, Portugal, Puerto Rico, Romania, Russia, Slovakia, Slovenia, South Africa, Spain, Spain, Sweden, Switzerland, Virgin Islands, Ukraine, United Kingdom, United States, Uruguay, Turkey, Japan.

Appendix……C: Plants as Material system: processing of hemp



Figure…..29: Traditional hemp processing

Field retting

Traditional

Hemp break



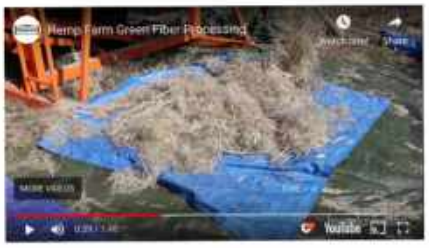
Scutching Hackling

Current

Decorticator









Figure…….30: Modern hemp processing

Appendix……C: Plants as Material system: East Yorkshire Hemp (Voase. 2019)



Figure……..31: East Yorkshire hemp, they produce many products namely construction products in this case



Figure……..32: East Yorkshire hemp, they produce many products construction products in this case

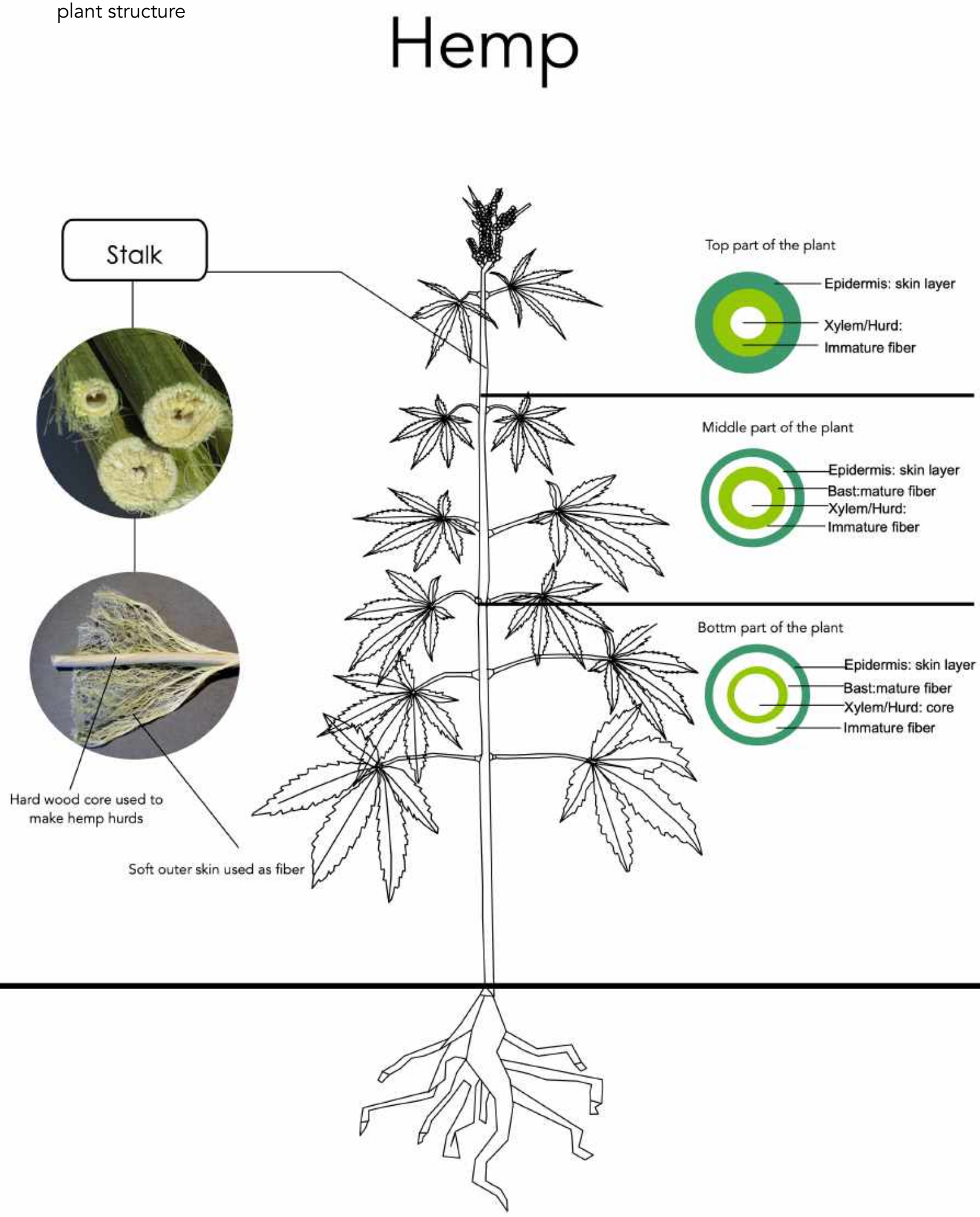


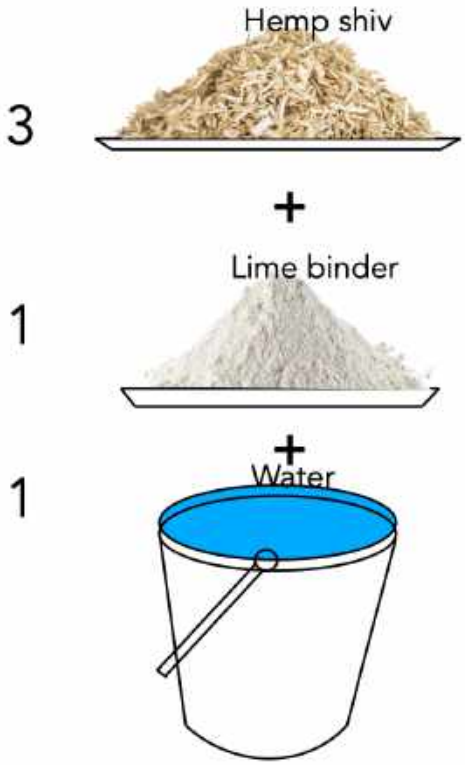
Figure……..33: Hemp being harvested 12 feet tall, every 3 Months



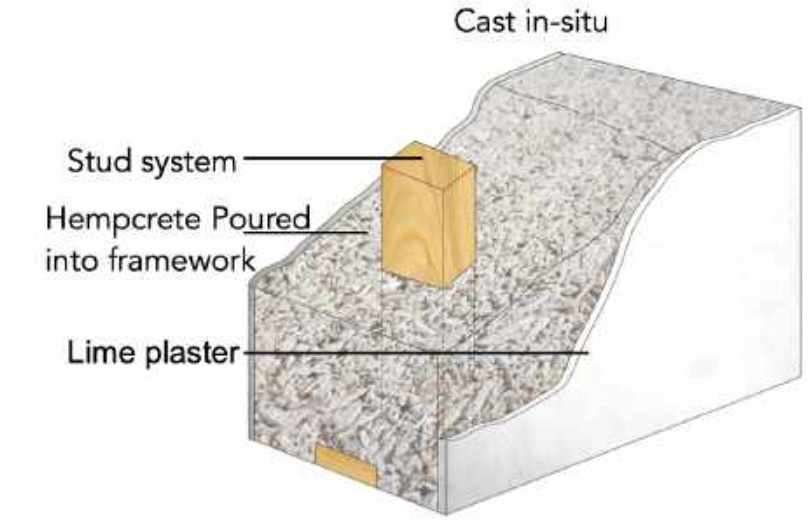
Figure……..34: Hemp field, 400 acres in size.

Figure……..35:
plant structure



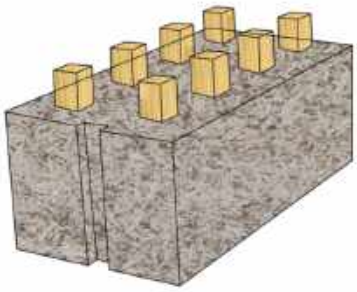


Figure……..36: Mixing ratio



Figure……..37: Cast in situ or pre-cast hempcrete

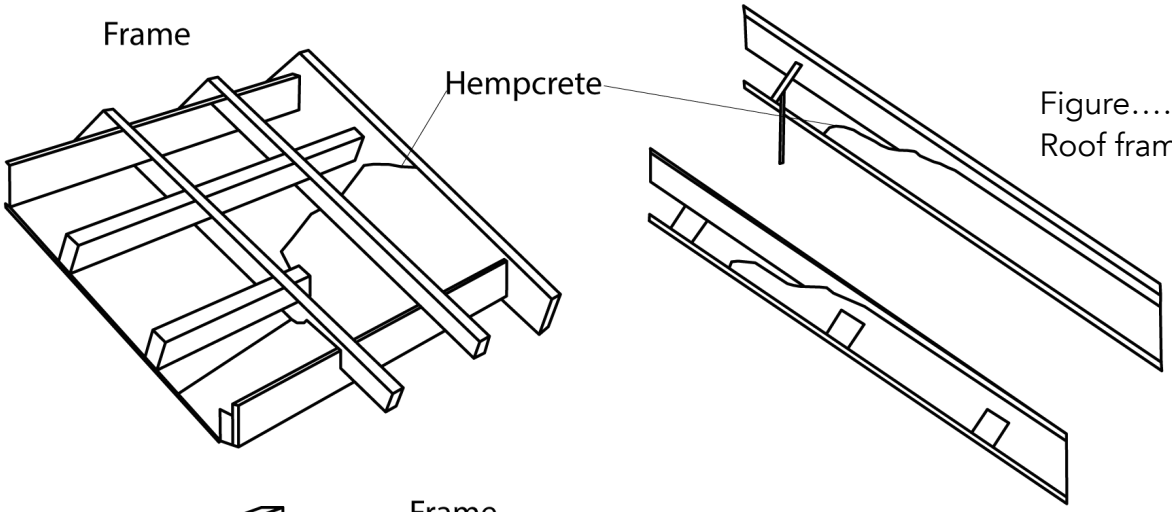
Pre-cast



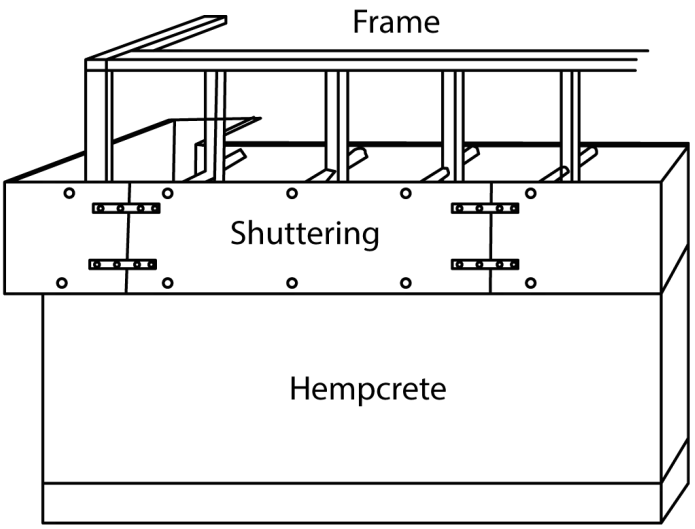
Cast into brick moulds with studs to join each brick or gaps so they can be put together



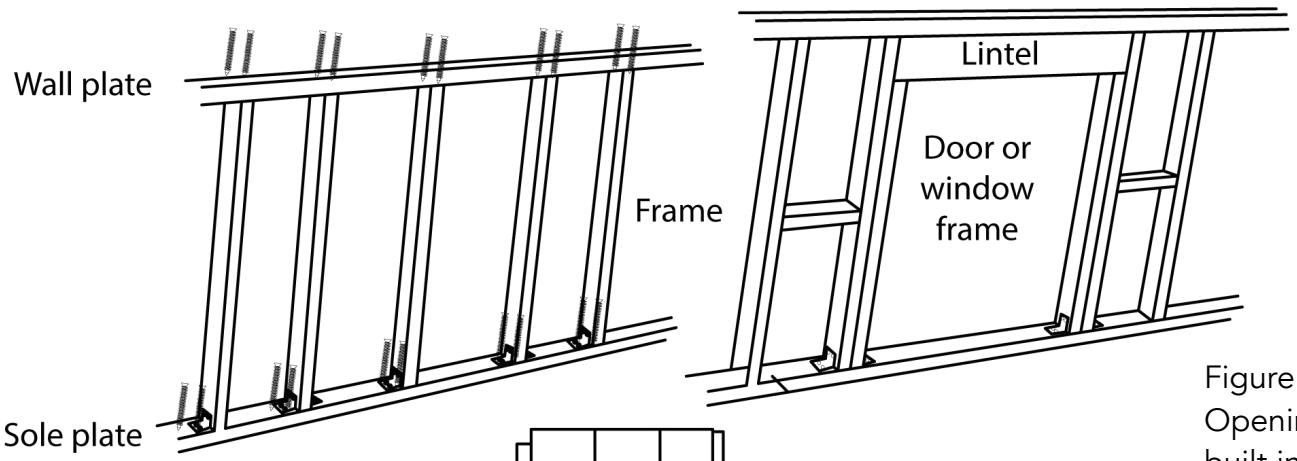
Hemp hurds composessed into



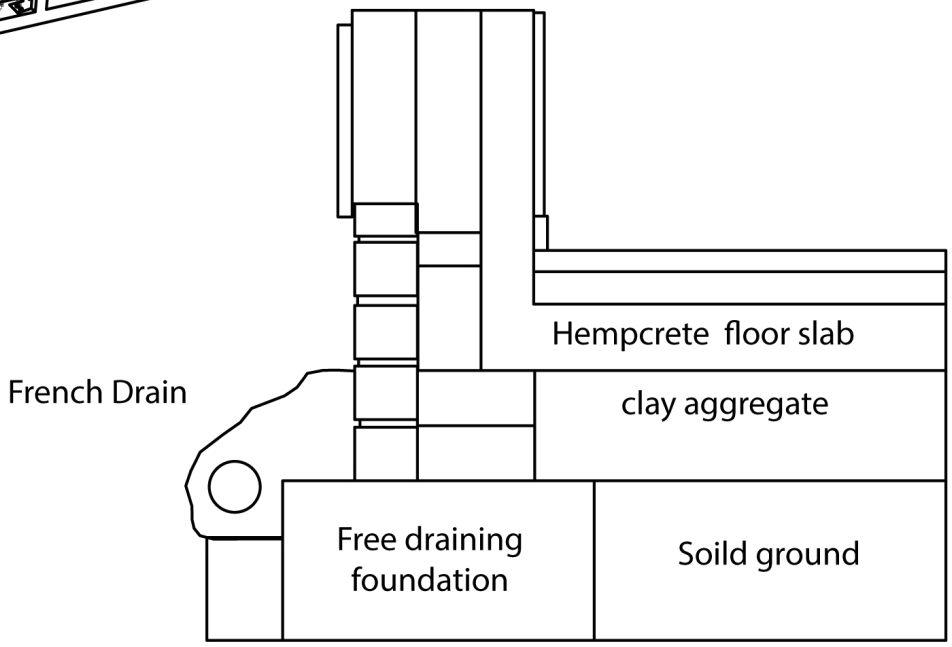
Figure……..38: Roof frame



Figure……..39: Shuttering to build the wall which is built in stages allowing it to dry

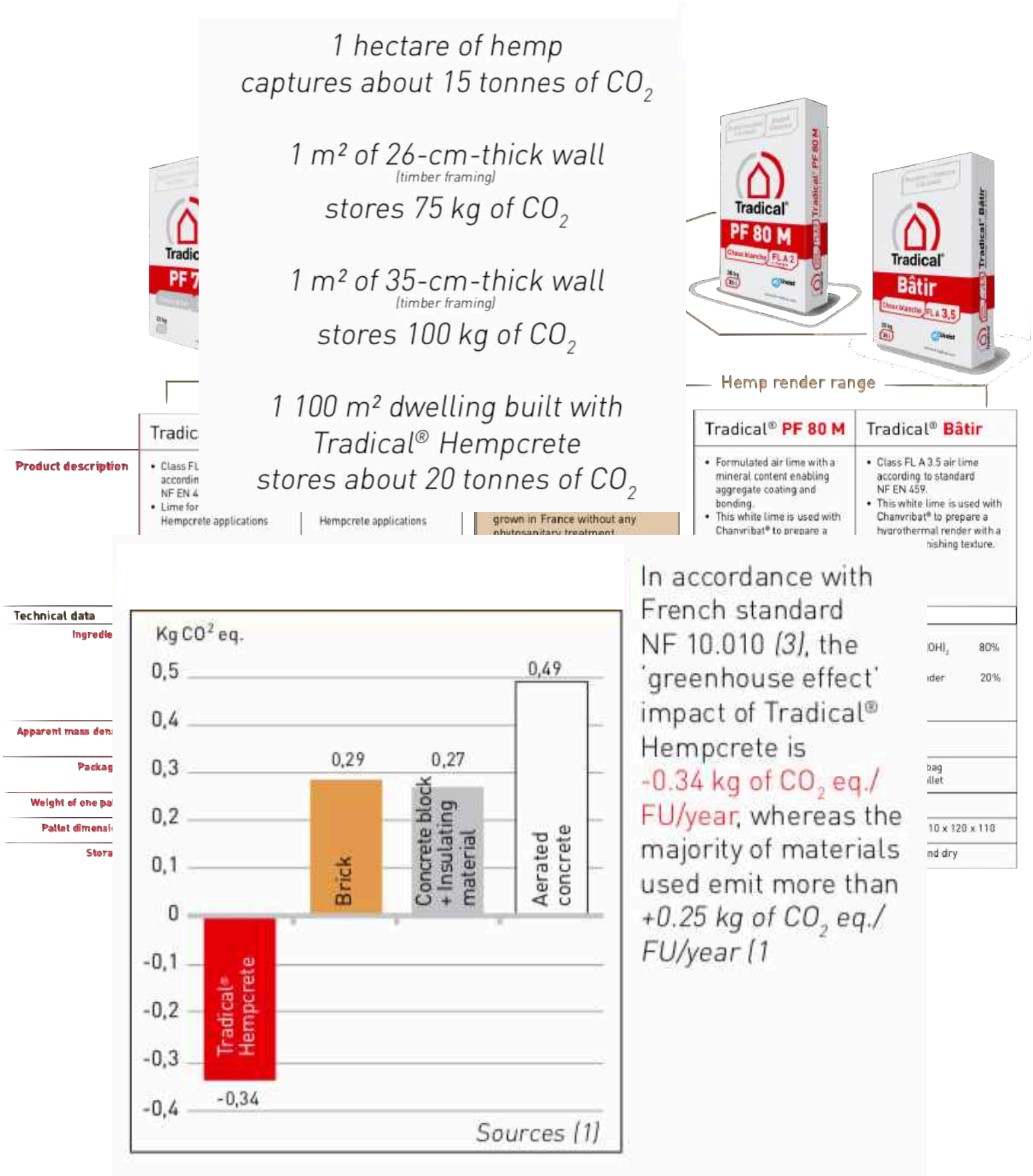


Figure……..40: Openings are built into the frame for doors and windows



Figure……..41: Floor slab with free drain and plinth to prevent continuous water exposure

Appendix……C: Tradical Hempcrete (TH)



Figure……43: Environmental benefits of using Tradical hempcrete.

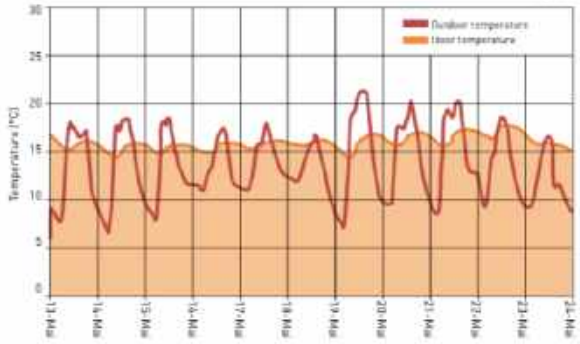
Appendix……C: Tradical Hempcrete (TH)

Absoprtion of high and low temperature variations

Comfort in WINTER

A wall with a thickness of 20 cm absorbs 80% of variation in outdoor temperature:

- for outdoor temperatures fluctuating between 6°C and 22°C over 24 hours, and for readings taken over 12 days in May, we can observe that the indoor temperature remains stable between 14°C and 17.5°C.

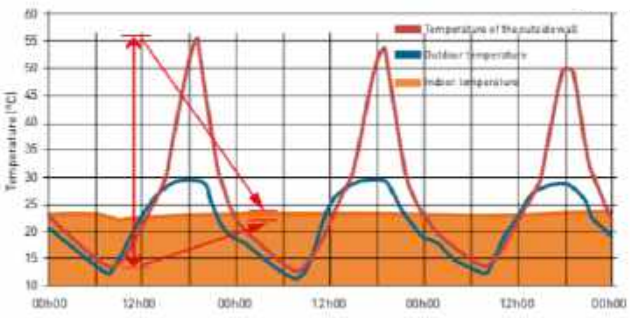


Humidity regulation

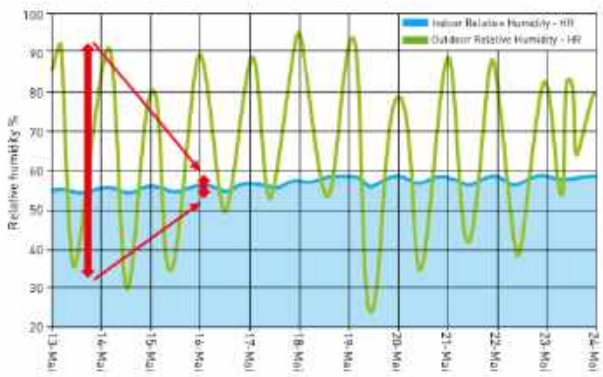
Comfort in SUMMER

A wall with a thickness of 30 cm absorbs 95% of variation in temperature on the surface of the outside wall:

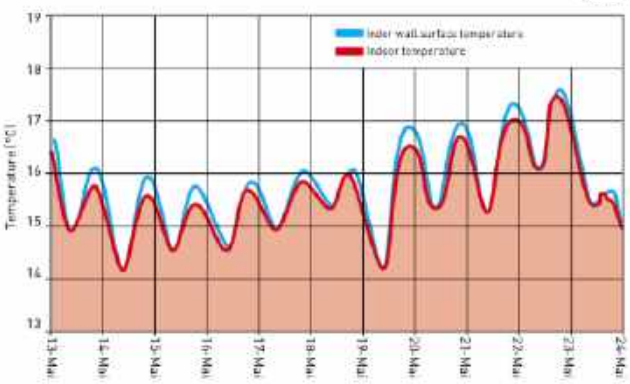
- for outdoor temperatures fluctuating between 12°C and 30°C, and for readings taken in September, we can observe that the indoor temperature remains stable between 22°C and 24°C.



Surface temperature regulation



Fire resistance



Acoustic performance

From the outset, Tradical Hempcretes have been subjected to numerous tests proving the effectiveness of this material in case of fire. For example, a wall with a thickness of 30 cm exposed to a temperature of 600°C does not increase in temperature, does not burn and does not emit smoke or particles.
These qualities make it excellent for protecting users or occupants as well as the building structure itself—especially timber-framed or half-timbered structures—whatever the architectural design (see pages 20 to 29).

Fire resistance classification

Tradical® Hempcretes - Walls
LNE (French laboratory) report for test no. P113132 DE/11
Rating = B - S1, d0

Tradical® Hempcretes - Roofs
LNE (French laboratory) report for test no. P113132 DE/8
Rating = B - S1, d0

Tradical® hygrothermal render
LNE (French laboratory) report for test no. P113132 DE/7
Rating = A2 - S1, d0

Tradical® Hempcretes also provide good acoustic performance ratings in restoration and construction situations, contributing to the reduction of indoor and outdoor noise.

Sound reduction index

For a 30-cm-thick Tradical® Hempcretes - Walls

Non-rendered exterior and interior walls BBRI (Belgian Building Research Institute) report for test no. AC 4956

R_w	C	C_{tr}
36 (-1;-5) dB	35.7 dB(A)	31.1 dB(A)

1 raw side - 1 side with 12.5 mm storey-high panel BBRI (Belgian Building Research Institute) report for test no. AC 4956

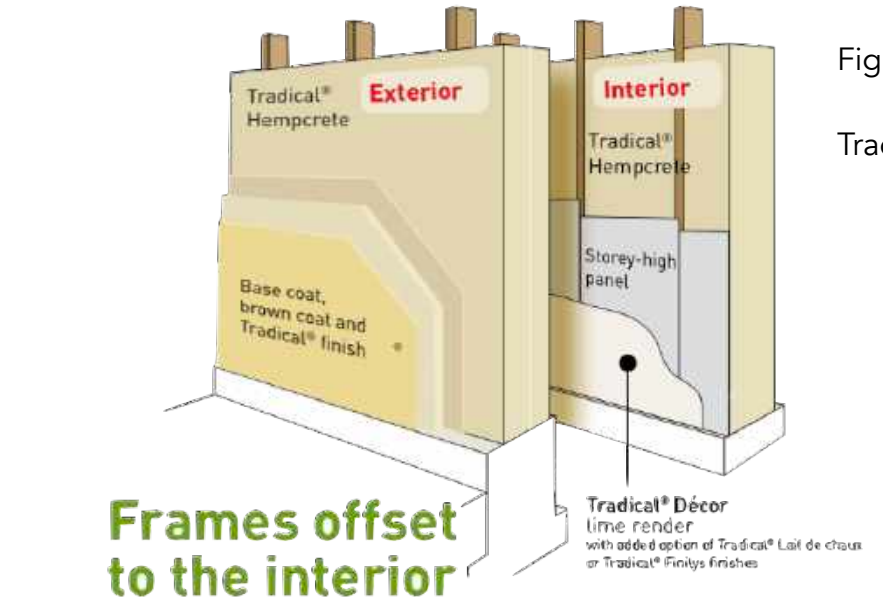
R_w	C	C_{tr}
49 (-1;-7) dB	48.5 dB(A)	42.5 dB(A)

Both sides covered with 12.5 mm storey-high panels BBRI (Belgian Building Research Institute) report for test no. AC 4955

R_w	C	C_{tr}
51 (-2;-7) dB	50 dB(A)	44 dB(A)

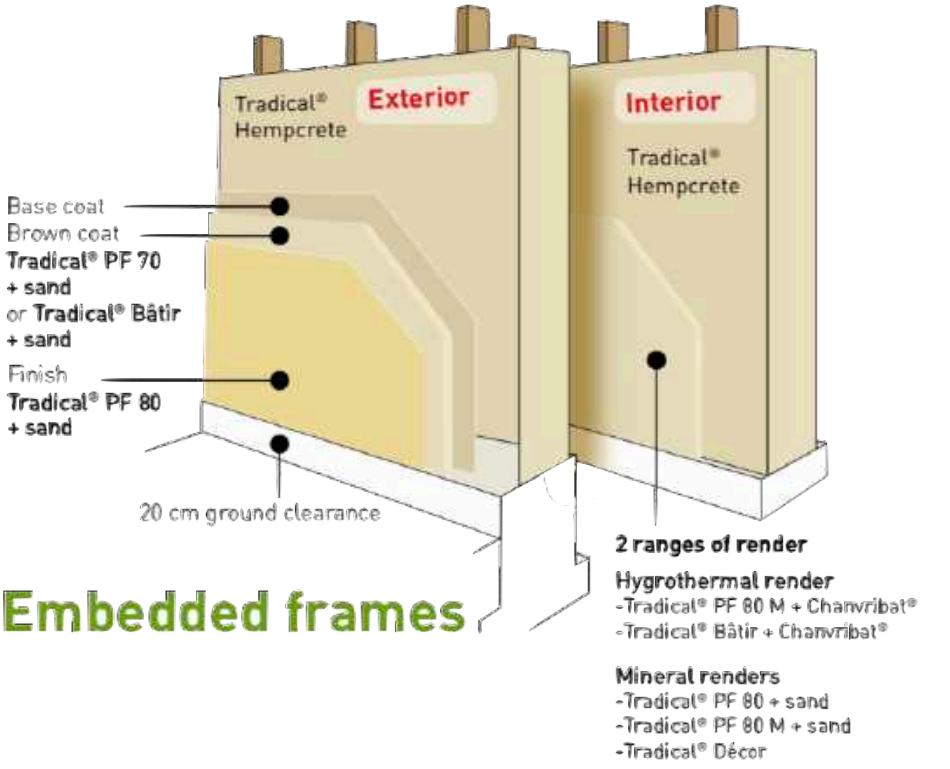
Figure……44: Performance benefits of using Tradical hempcrete.

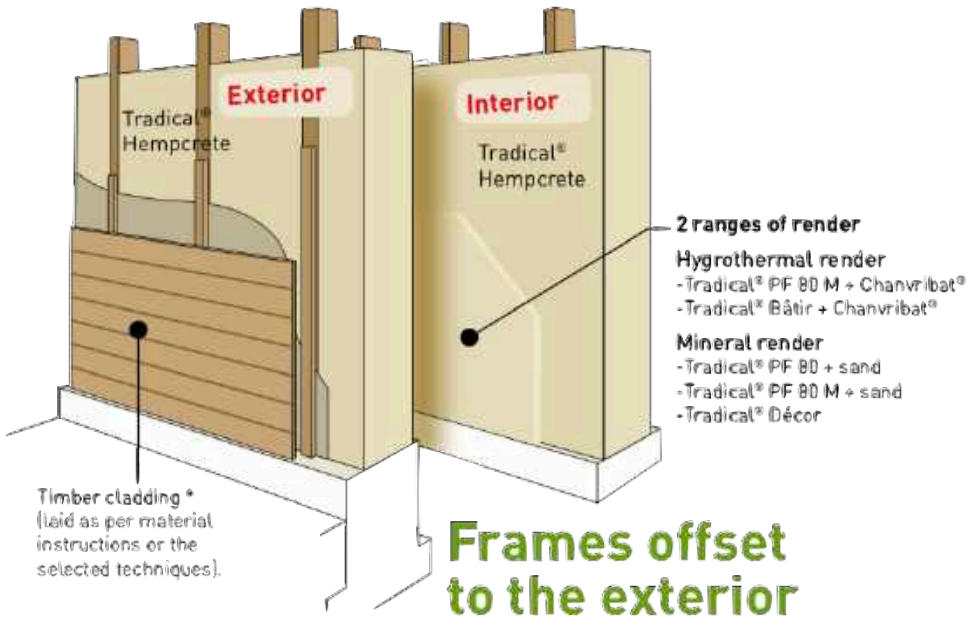
Appendix……C: Tradical Hempcrete (TH)



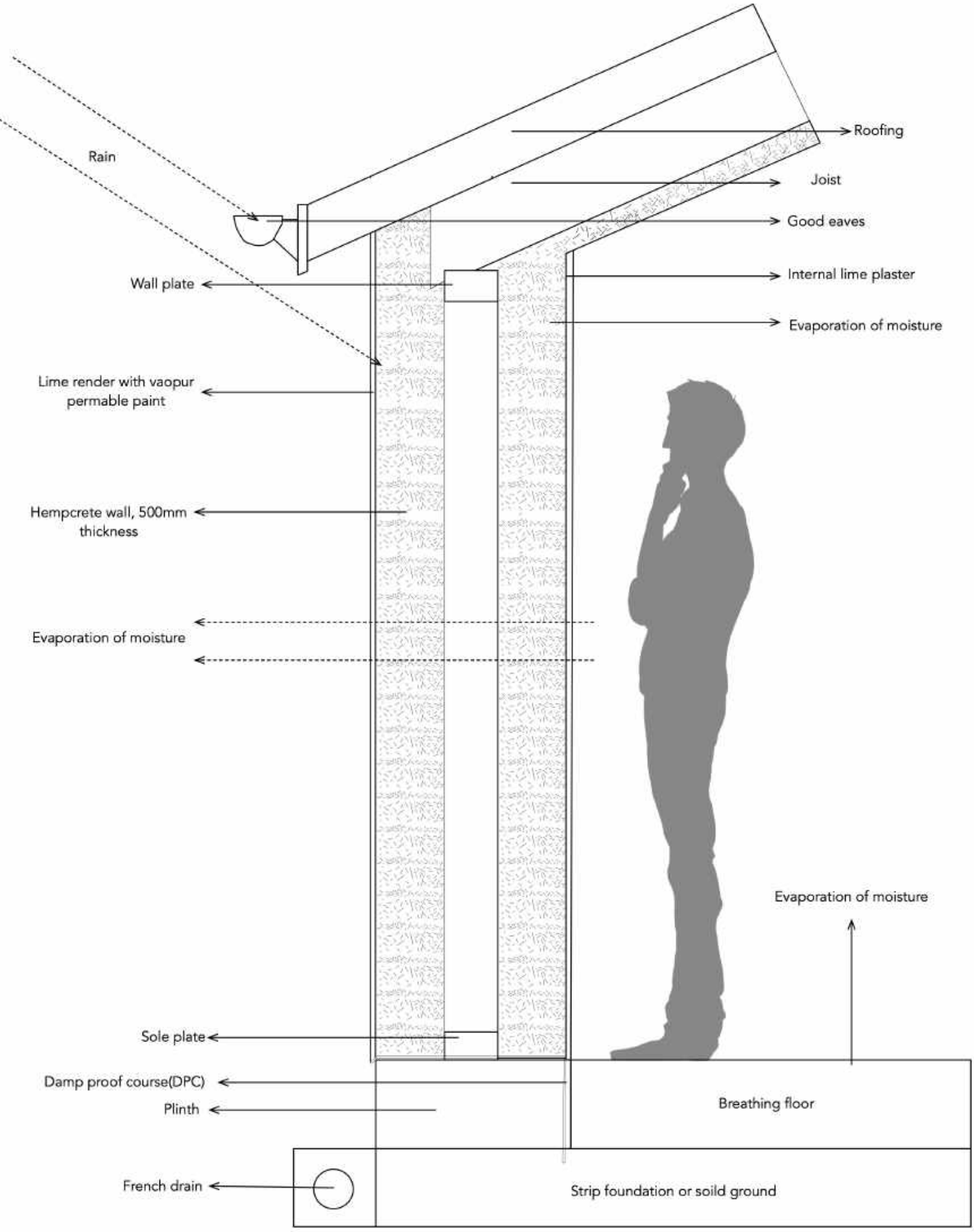
Figure……..45:

Tradical wall systems

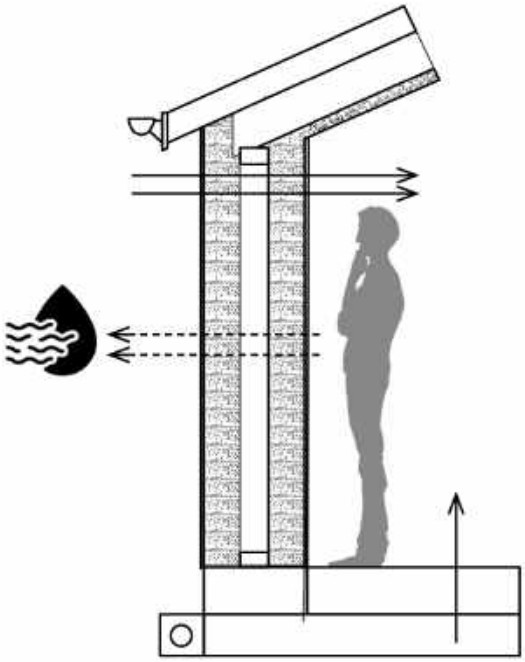




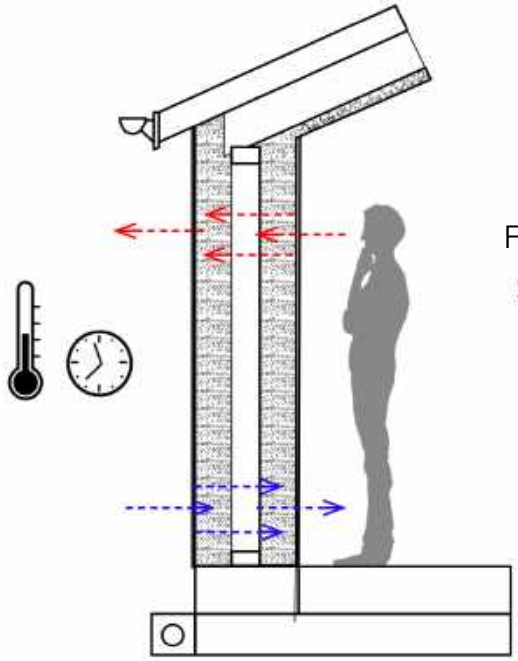
Appendix……C: Breathable wall



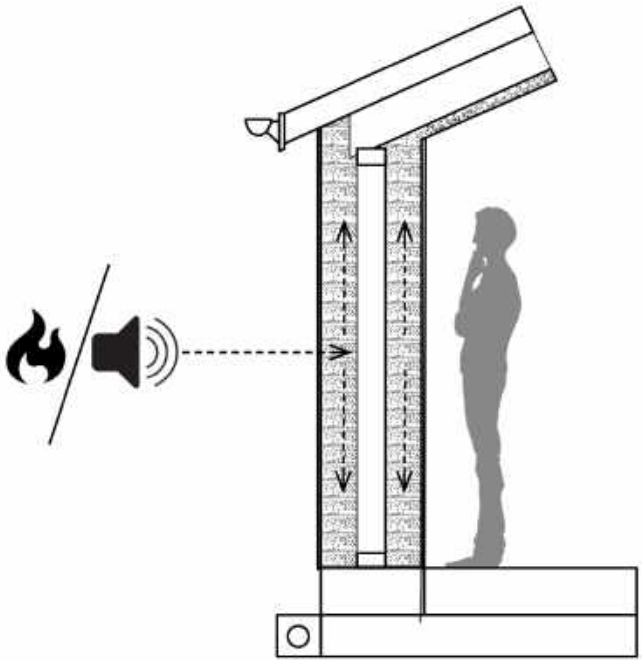
Figure……..46: Breathable wall works by allowing evaporation between the hurts and the binder within the wall, the hygroscopic nature leads to moisture control which ensures no mold build up occurs



Figure……..47:
Humidity regulation



Figure……..48:
Slow release of hot and cold air



Figure……..49:
Sound and fire abroption

Appendix……C: Wise building case study



Figure……..50: Courtyard used for collecting water



Figure……..51: Internal curved walkways



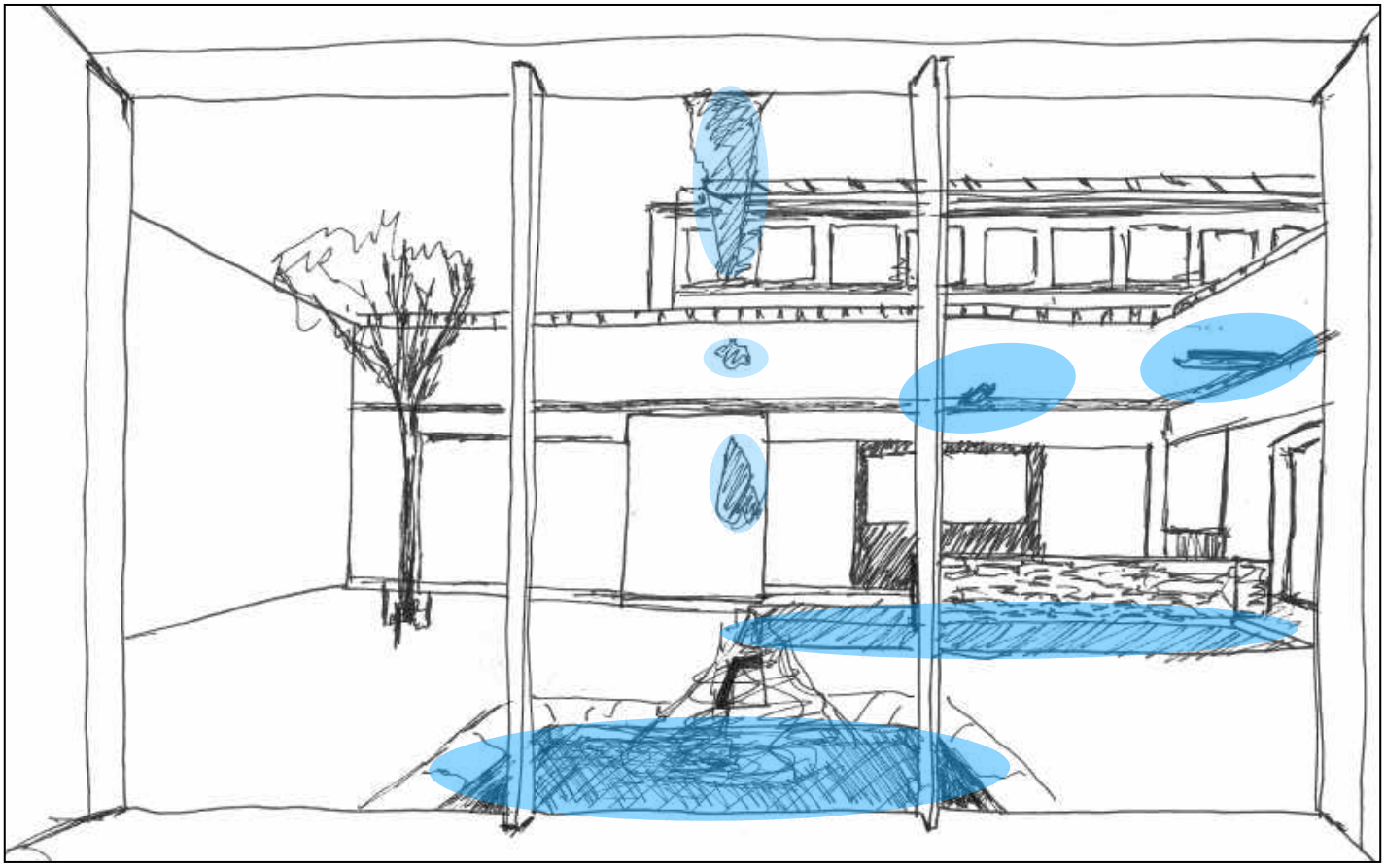
Figure……..52: Shepard theatre

The sunlight roof can be opened and closed according to the day

Appendix……C: Wise building case study



Figure……..53: Court yard view from inside



Figure……..54: Water collection from roof using cascades

Appendix……C: Wise building case study

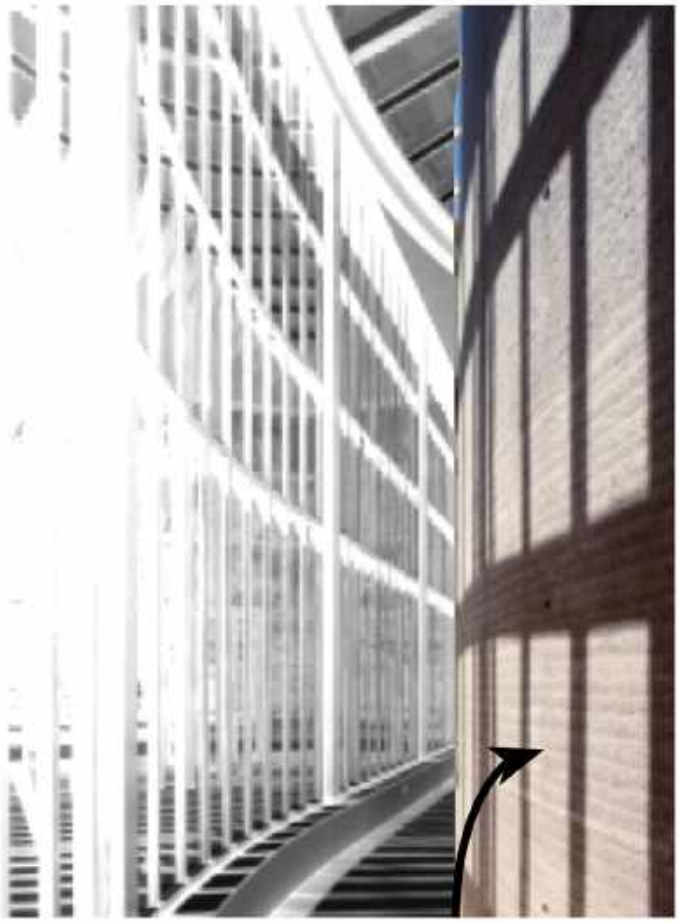
Figure……..56

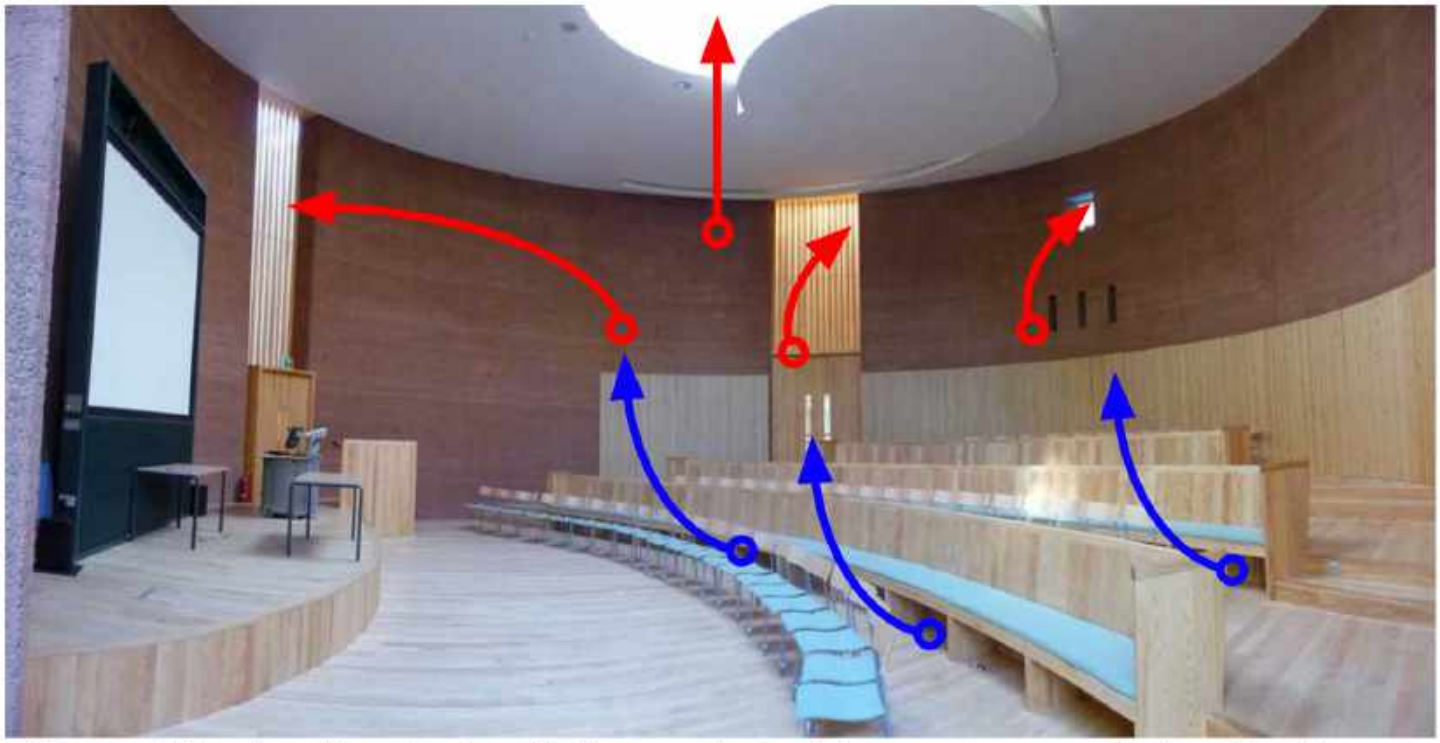
Top light for solar gain and ventiltation



Figure……..57

Curved walls

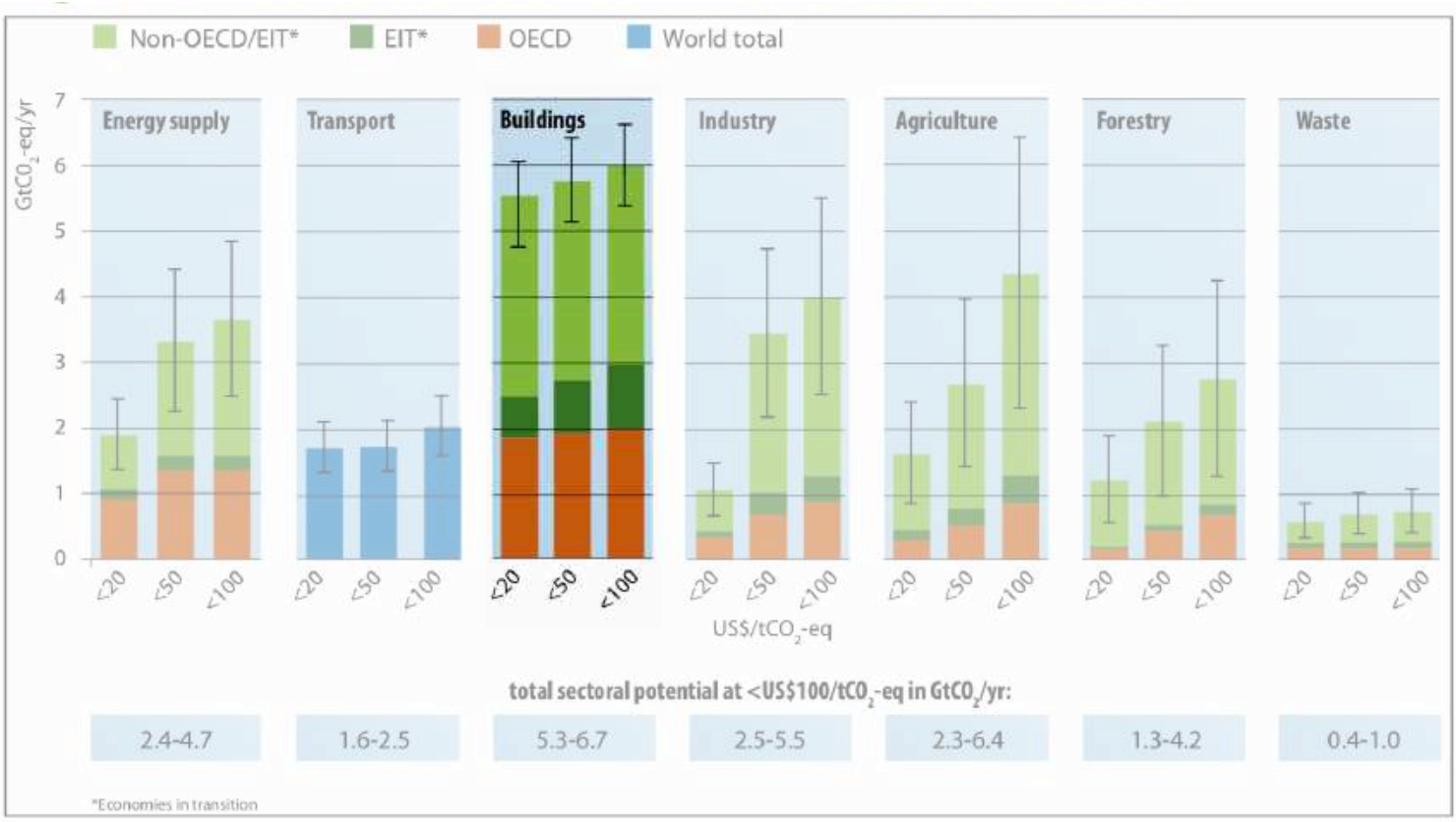




Sheprard threate with natural veniltation constrols. A large open cruved lecture thearte.

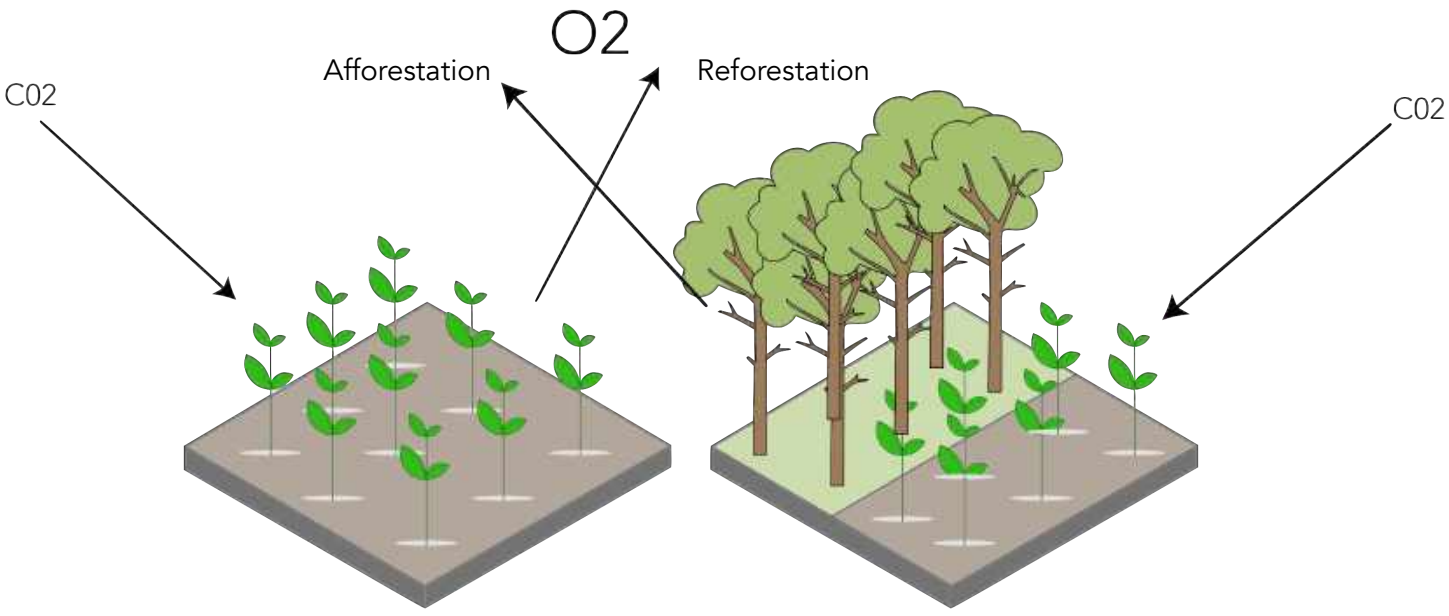
Figure……..58

Appendix……D: Environmental and socio-economic benefits



Figure……..59: CO2 mitigation potential

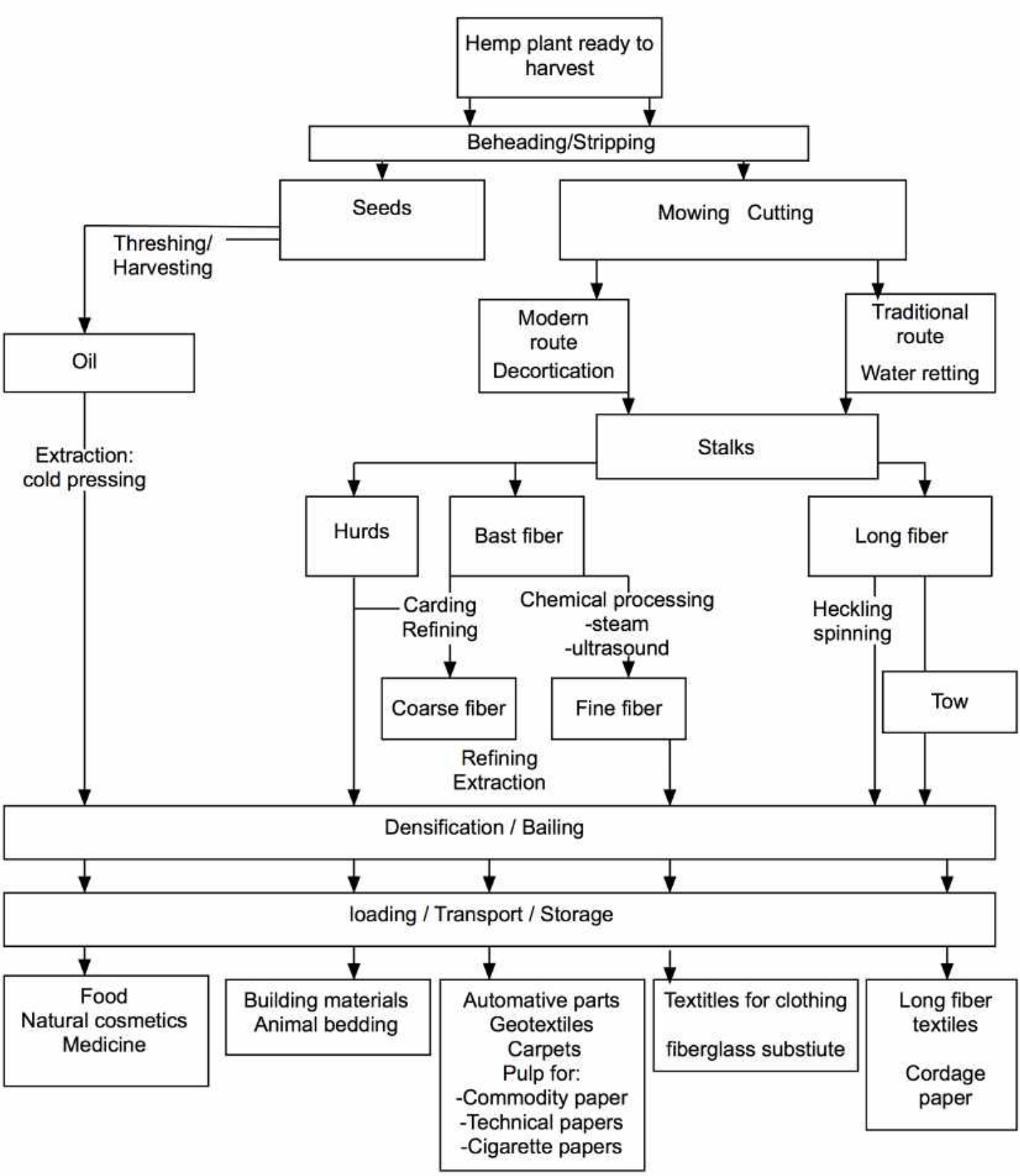
Building sector CO2 mitigation potential



Figure……..60: Carbon sink: carbon sequestration.
Afforestation(planting trees) and reforestation (planting trees where previous ones once were)

(IPCC, 2018)

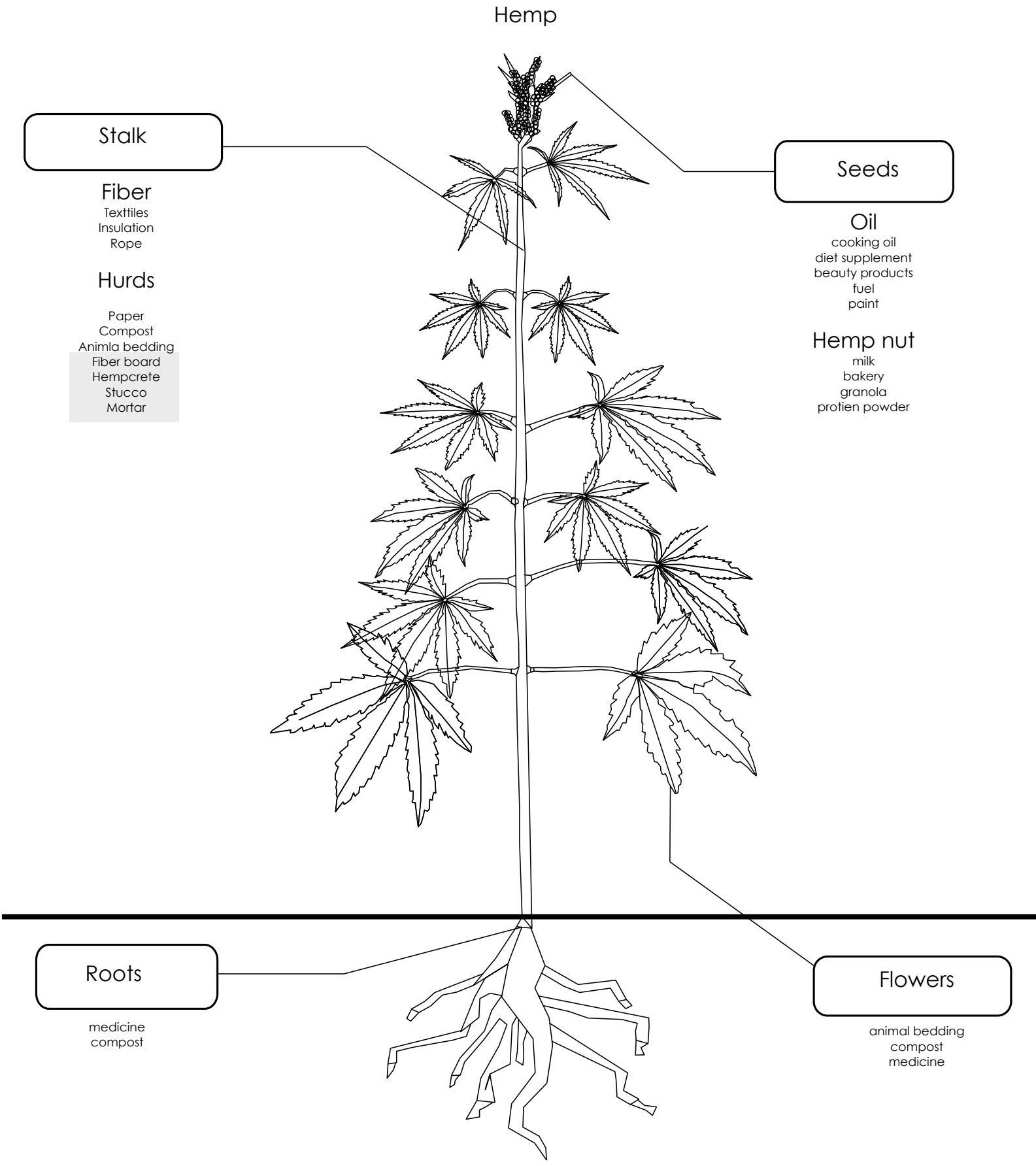
Appendix……D: Environmental and socio-economic benefits



Figure……..61: Hemp supply chain

The long supply chain suggests a large workforce is needed

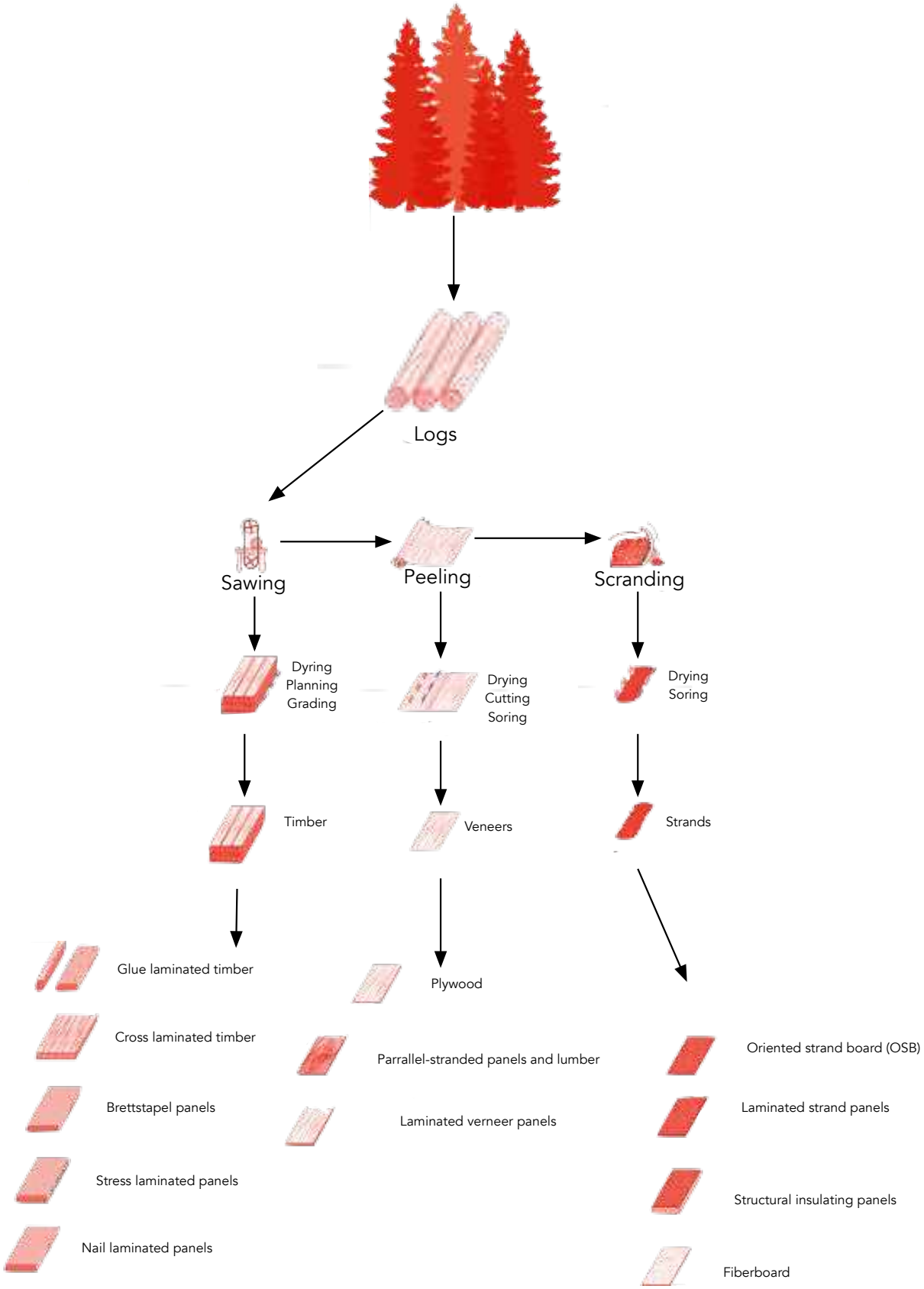
Appendix……D: Environmental and socio-economic benefits



Figure……..62: The uses of hemp

The varied uses suggest a number of jobs will be created, 25 000 products to be exact, (Johnson, 2018, p. 1)

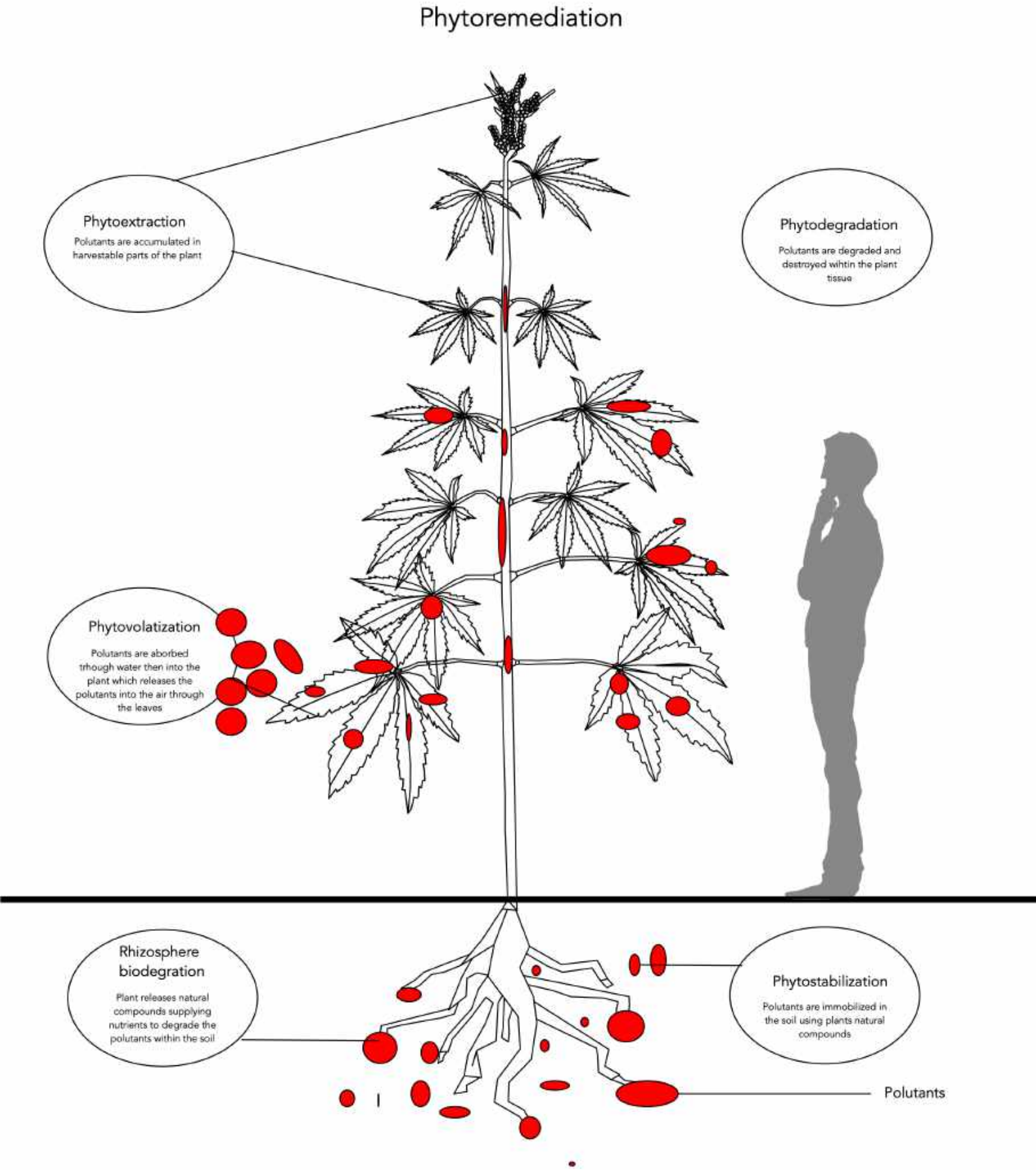
Appendix……D: Environmental and socio-economic benefits



Figure……..63: Timber supply chain

The supply chain suggests a large workforce is needed

Appendix……D: Environmental and socio-economic



Figure……..64: Phytoremediation is a process where plants absorb pollutants within the soil and groundwater and transfer it to the

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2019 – 2020

This dissertation examined plants as a structural and material system, using Frei Otto's branching precedents and hempcrete case studies to inform a combined structural and material strategy. The research concludes that combining plant-derived structural and material systems can produce sustainable, renewable buildings with improved performance and reduced environmental impact, directly informing the design thesis that followed.



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